Errata and Source Corrections for the NIST Digital Library of Mathematical Functions

found in the Lean 4 formalization `lean-DLMF`

Bastiaan J. Braams

2026-10-26

Abstract

This report collects, project-wide, every place where a formula of the NIST Digital Library of Mathematical Functions (DLMF) [1] could not be formalized exactly as printed, together with the earlier defects in the project's own transcriptions that were found and fixed. It is meant for readers who compare a Lean statement with the printed DLMF formula and find that the two differ, and as a list of candidate errata reports for the DLMF editors. The entries refer to DLMF Release 1.2.8 of 2026-09-15 and describe the state of the `lean-DLMF` project on 2026-10-26. The report has 97 entries of genuine DLMF errata, 23 project transcription defects that were fixed, and 43 formalization notes.

Contents

1 Introduction **2**
1.1 The project . . . 2
1.2 What is recorded . . . 2
1.3 How the entries were found and verified . . . 2
1.4 Counts . . . 3

2 (a) Genuine DLMF errata **3**
2.1 Chapter 4 (elementary functions) . . . 3
2.2 Chapter 8 (incomplete Gamma and related functions) . . . 5
2.3 Chapter 9 (Airy functions) . . . 6
2.4 Chapter 11 (Struve and related functions) . . . 6
2.5 Chapter 12 (parabolic cylinder functions) . . . 7
2.6 Chapter 13 (confluent hypergeometric functions) . . . 7
2.7 Chapter 14 (Legendre and related functions) . . . 8
2.8 Chapters 16, 17 and 24 . . . 9
2.9 Chapter 19 (elliptic integrals) . . . 11
2.10 Chapter 20 (theta functions) . . . 15
2.11 Chapter 21 (Riemann theta functions) . . . 16
2.12 Chapter 22 (Jacobian elliptic functions) . . . 17
2.13 Chapter 23 (Weierstrass elliptic and modular functions) . . . 18
2.14 Chapter 24 (Bernoulli and Euler polynomials) . . . 21
2.15 Chapter 25 (zeta and related functions) . . . 22
2.16 Chapter 26 (combinatorial analysis) . . . 24
2.17 Chapter 27 (functions of number theory) . . . 27
2.18 Chapters 28–31 (Mathieu, Lamé, spheroidal and Heun functions) . . . 28

3 (b) Project transcription defects that were fixed **31**

4 (c) Formalization notes **34**
4.1 Totalization, removable points and nonvanishing hypotheses 34
4.2 Branches and multivalued sets 37
4.3 Implicit conventions made explicit 38
4.4 Chapters 24–27 38
4.5 Chapters 28–31 39
4.6 DLMF's own published errata followed 40

5 Outside the DLMF scope **40**

1 Introduction

1.1 The project

`lean-DLMF` is a formalization in Lean 4, on top of Mathlib, of special functions following the NIST Digital Library of Mathematical Functions [1]. It began as a formalization of *Special Functions* by Andrews, Askey and Roy [3] (henceforth AAR); the DLMF chapters are now the organizing principle, with AAR, Temme, Beals–Wong [4] and Duverney as supporting sources. The aim is to identify gaps in the special-functions coverage of Mathlib and to fill them. Each formalized DLMF statement is tagged with its DLMF equation number; a statement whose printed form had to be changed is recorded in this report.

1.2 What is recorded

Entries are grouped in three categories.

(a) Genuine DLMF errata (Section 2): the printed formula is false, or lacks a condition it needs, under DLMF's own definitions and conventions.

(b) Project transcription defects that were fixed (Section 3): DLMF is right, but an earlier Lean statement or definition in this project was wrong (wrong bracket, sign, binder, domain or set reading).

(c) Formalization notes (Section 4): DLMF is right in its own reading, but the Lean statement needs an extra hypothesis or a separate case because of Lean's conventions ($x/0 = 0$, $\Gamma(0) = 0$, totalized $\zeta(1)$, unordered sums), a branch choice, a removable point, or an implicit convention made explicit. DLMF's own published errata [2] that the formalization follows are also listed here.

Section 5 ("Outside the DLMF scope") records errors found in the supporting sources (AAR and Beals–Wong), together with two DLMF items that were checked and cleared.

Each entry gives the DLMF identifier, linked to the DLMF web page (equation numbers are cited as, for example, DLMF (28.28.16)); what is printed against what is correct, with the printed formula taken from DLMF's own LaTeX source where a formula is involved; the evidence; the Lean declaration(s) and the file that holds them; the status (proved, admitted, definition, or an upstream pointer to Mathlib); and the date found. "Admitted" means that the corrected statement is in Lean with a `sorry` proof. Some entries group closely related equation numbers. File names such as `REVIEW.md` and `DLMF19Acquisition.md` refer to the review log and the chapter acquisition ledgers of the project repository.

1.3 How the entries were found and verified

For entries found before 2026-10-25 the evidence is summarized from the cited section of `REVIEW.md` or a chapter ledger and was not re-verified for this register. The Chapter 19 entries

were checked the same way (see the introduction to Section 2.9). The Chapter 20–23 entries were found by the acquisition agents and confirmed by an independent numerical check (`REVIEW.md`, section "Chapters 20–23 acquisition (2026-10-25)", subsection "DLMF errata and source corrections"; ledgers `DLMF20Acquisition.md`–`DLMF23Acquisition.md`). The Chapter 24–27 entries were found by the acquisition agents and confirmed by an independent numerical and brute-force check (`REVIEW.md`, section "Chapters 24–27 acquisition (2026-10-25)"; ledgers `DLMF24Acquisition.md`–`DLMF27Acquisition.md`); three suspected items that the check did not confirm are listed under (c) or dropped. The Chapter 28–31 entries were found by the acquisition agents on 2026-10-26 and re-derived by an independent verifier from the DLMF 1.2.8 page text with its own numerics.

1.4 Counts

This version has 97 entries in (a), 23 in (b) and 43 in (c), and 3 entries outside the DLMF scope; some entries group closely related equation numbers. Entries are numbered a.1, a.2, …, b.1, …, c.1, … within each category.

2 (a) Genuine DLMF errata

2.1 Chapter 4 (elementary functions)

a.1 (4.7.13) (also §4.7(ii), paragraph 2)

Printed the general solution of the linear first-order equation $\mathrm{d}w/\mathrm{d}z = f(z)w$ (4.7.12) is given as

$$w = \exp\left(\int f(z)\,\mathrm{d}z\right) + \text{constant}.$$

Correct a multiplicative constant:

$$w = C\exp\left(\int f(z)\,\mathrm{d}z\right).$$

Evidence proved in Lean.

Lean `DLMF.Elementary.linear_ode_solution` (`DLMF/Elementary/LogExpAnalysis.lean`), proved.

Found 2026-10-24 (`DLMF4Acquisition.md`, "Domains, conventions and source corrections").

a.2 (4.8.2)

Printed

$$\ln(z_1 z_2) = \ln z_1 + \ln z_2, \qquad -\pi \le \operatorname{ph} z_1 + \operatorname{ph} z_2 \le \pi.$$

The identity fails at the endpoint $-\pi$: $z_1 = z_2 = -\mathrm{i}$ gives $\mathrm{i}\pi$ against $-\mathrm{i}\pi$.

Correct

$$\ln(z_1 z_2) = \ln z_1 + \ln z_2, \qquad -\pi < \operatorname{ph} z_1 + \operatorname{ph} z_2 \le \pi.$$

Lean upstream pointer to Mathlib `Complex.log_mul` (`DLMF/Elementary/MathlibCoverage.lean`).

Found 2026-10-21 (`REVIEW.md`, "Upstream pointers (2026-10-21)").

a.3 (4.8.9)

Printed

$$\ln(\exp z) = z, \qquad -\pi \le \Im z \le \pi.$$

This fails at $\Im z = -\pi$ ($z = -\mathrm{i}\pi$ gives $\mathrm{i}\pi$).

Correct

$$\ln(\exp z) = z, \qquad -\pi < \Im z \le \pi.$$

Lean upstream pointer to Mathlib `Complex.log_exp`.

Found 2026-10-21 (as (4.8.2)).

a.4 (4.20.13), (4.20.14) (borderline (a)/(c): wording of "general solution")

Printed with $a \neq 0$, the general solutions of $(\mathrm{d}w/\mathrm{d}z)^2 + a^2w^2 = 1$ (4.20.10) and $\mathrm{d}w/\mathrm{d}z - a^2w^2 = 1$ (4.20.11) are given as

$$w = (1/a)\sin(az + c), \qquad w = (1/a)\tan(az + c),$$

where c is an arbitrary constant.

Correct these "general solutions" omit the constant solutions $w = \pm 1/a$ (for (4.20.13)) and $w = \pm \mathrm{i}/a$ (for (4.20.14)).

Lean `trig_energy_ode_general`, `trig_riccati_ode_general` (`DLMF/Elementary/Trigonometric.lean`), proved.

Found 2026-10-24 (`DLMF4Acquisition.md`, 4.20 rows).

a.5 (4.26.9), (4.26.10)

Printed for integers m, n,

$$\int_0^{\pi} \sin(mt)\sin(nt)\,\mathrm{d}t = 0, \qquad \int_0^{\pi} \cos(mt)\cos(nt)\,\mathrm{d}t = 0, \qquad m \neq n.$$

The integer-frequency orthogonality is false at $m = -n$.

Correct the condition must be $m \neq n$ and $m \neq -n$.

Lean `integral_sin_integer_orthogonal`, `integral_cos_integer_orthogonal` (`DLMF/Elementary/Trigonometric.lean`), proved.

Found 2026-10-24 (`DLMF4Acquisition.md`, 4.26 rows).

a.6 (4.34.12), (4.34.13), (4.34.14) (borderline (a)/(c), as (4.20.13)–(4.20.14))

Printed with $a \neq 0$, the general solutions of $(\mathrm{d}w/\mathrm{d}z)^2 - a^2w^2 = 1$, $(\mathrm{d}w/\mathrm{d}z)^2 - a^2w^2 = -1$ and $\mathrm{d}w/\mathrm{d}z + a^2w^2 = 1$ (4.34.8)–(4.34.10) are given as

$$w = (1/a)\sinh(az + c), \qquad w = (1/a)\cosh(az + c), \qquad w = (1/a)\coth(az + c),$$

where c is an arbitrary constant.

Correct these "general solutions" omit the constant equilibria (for (4.34.14): $w = \pm 1/a$).

Lean `hyperbolic_squared_ode_plus`, `hyperbolic_squared_ode_minus`, `hyperbolic_riccati_ode` (`DLMF/Elementary/Hyperbolic.lean`), admitted.

Found 2026-10-24 (`DLMF4Acquisition.md`, 4.34 rows).

a.7 (4.37.11)

Printed the sign pairing

$$\operatorname{arccosh}(-z) = \pm\pi\mathrm{i} + \operatorname{arccosh} z, \qquad \Im z \gtrless 0,$$

that is, $\operatorname{arccosh}(-z) = \operatorname{arccosh}(z) + \operatorname{sgn}(\Im z)\,\pi\mathrm{i}$.

Correct

$$\operatorname{arccosh}(-z) = \operatorname{arccosh} z - \operatorname{sgn}(\Im z)\,\pi\mathrm{i}.$$

Evidence DLMF (4.37.20) gives $\operatorname{arccosh}(\mathrm{i}) = \ln(1+\sqrt{2}) + \pi\mathrm{i}/2$ and $\operatorname{arccosh}(-\mathrm{i}) = \ln(1+\sqrt{2}) - \pi\mathrm{i}/2$, contradicting the printed sign; proved in Lean.

Lean `DLMF.Elementary.arccosh_neg` (`DLMF/Elementary/Hyperbolic.lean`), proved.

Found 2026-10-24 (`DLMF4Acquisition.md`, 4.37.E11 row).

a.8 (4.37.14)

Printed the reciprocal of (4.37.11):

$$\operatorname{arcsech}(-z) = \mp\pi\mathrm{i} + \operatorname{arcsech} z, \qquad \Im z \gtrless 0.$$

The printed pairing is reversed.

Correct

$$\operatorname{arcsech}(-z) = \operatorname{arcsech} z + \operatorname{sgn}(\Im z)\,\pi\mathrm{i}.$$

Lean `DLMF.Elementary.arcsech_neg` (`DLMF/Elementary/Hyperbolic.lean`), proved.

Found 2026-10-24.

2.2 Chapter 8 (incomplete Gamma and related functions)

a.9 (8.10.11)

Printed

$$(1-\mathrm{e}^{-\alpha_a x})^a \le P(a,x) \le (1-\mathrm{e}^{-\beta_a x})^a, \qquad x \ge 0,\ a > 0,$$

followed by "Equalities in (8.10.11) apply only when $a = 1$."

Correct the statement about equality also needs $x > 0$: at $x = 0$ both bounds are 0 for all a.

Lean `normIncGammaP_exponential_bounds`, `normIncGammaP_exponential_bounds_strict` (`DLMF/IncompleteGamma/BoundsFractions.lean`), proved.

Found 2026-10-19 (`DLMF678Acquisition.md`, "Source qualifications and a proved counterexample").

a.10 §8.17 (claim after (8.17.23))

Printed for the incomplete-Beta continued fraction

$$I_x(a,b) = \frac{x^a(1-x)^b}{a\,\mathrm{B}(a,b)} \cfrac{1}{1+\cfrac{d_1}{1+\cfrac{d_2}{1+\cfrac{d_3}{1+\cdots}}}}$$

DLMF (8.17.22), DLMF states: "The $4m$ and $4m+1$ convergents are less than $I_x(a,b)$, and the $4m+2$ and $4m+3$ convergents are greater than $I_x(a,b)$", i.e. the convergents alternate as upper and lower bounds.

Correct the claim is false.

Evidence $a = 1$, $b = 1/2$, $x = 1/2$: $I = 1 - \sqrt{2}/2$, while the two-level convergent $\sqrt{2}/5$ and the three-level convergent $23\sqrt{2}/112$ are both below I; proved in Lean.

Lean `incBeta_cf_upper_bound_counterexample`, `incBeta_cf_third_bound_counterexample` and value lemmas (`DLMF/IncompleteGamma/Beta.lean`), proved.

Found 2026-10-19 (`DLMF678Acquisition.md`, same section).

a.11 (8.19.25)

Printed

$$\int_0^\infty \mathrm{e}^{-at} t^{b-1} E_p(t)\,\mathrm{d}t = \frac{\Gamma(b)\,(1+a)^{-b}}{p+b-1}\,F\left(1,b;p+b;a/(1+a)\right),$$

with the conditions $\Re a > -1$, $\Re(p+b) > 1$.

Correct the printed conditions are insufficient: $\Re b > 0$ is needed as well, for convergence at the zero endpoint.

Lean `genExpIntE_mellin_laplace` (`DLMF/IncompleteGamma/ExponentialIntegral.lean`), proved.

Found 2026-10-19 (`DLMF678Acquisition.md`, same section).

2.3 Chapter 9 (Airy functions)

a.12 (9.10.7)

Printed as $x \to -\infty$,

$$\int_{-\infty}^{x} \mathrm{Bi}\,(t)\,\mathrm{d}t = \pi^{-1/2}(-x)^{-3/4}\sin\left(\tfrac{2}{3}(-x)^{3/2}+\tfrac{1}{4}\pi\right)+O\left(|x|^{-9/4}\right).$$

Correct the leading term has the sign $-$:

$$\int_{-\infty}^{x} \mathrm{Bi}\,(t)\,\mathrm{d}t = -\pi^{-1/2}(-x)^{-3/4}\sin\left(\tfrac{2}{3}(-x)^{3/2}+\tfrac{1}{4}\pi\right)+O\left(|x|^{-9/4}\right).$$

Evidence at $x=-10$ the integral is -0.01504 against the printed $+0.01240$; the derivative of the printed term is $-\mathrm{Bi}(x)$; proved in Lean.

Lean `airyBi_negative_integral_asymptotic` (`DLMF/Airy/Integrals.lean`), proved.

Found 2026-10-21 (`REVIEW.md`, "Round 7 statement corrections (2026-10-21, user-approved)").

a.13 (9.13.36)

Printed for all integer p,

$$B_3\,(z,p)-B_1\,(z,p)=2\pi\mathrm{i}A_2\,(z,p)\,.$$

Correct

$$B_3\,(z,p)-B_1\,(z,p)=2\pi\mathrm{i}(A_2\,(z,p)+B_0\,(z,p));$$

the printed form holds only for $p\le 0$.

Evidence the failure at $p=1$ is proved in Lean (`reidAiry_connection_36_ne_one`).

Lean `reidAiry_connection_36`, `reidAiry_connection_36_of_nonpos`, `reidAiry_connection_36_ne_one` (`DLMF/Airy/GeneralizedContours.lean`), proved.

Found 2026-10-20 (`REVIEW.md`, "Shared-helper round (priority 2, 2026-10-20)").

2.4 Chapter 11 (Struve and related functions)

a.14 (11.4.15)

Printed for the relative error ϑ in (11.4.14),

$$|\vartheta|<\frac{2}{3}\exp\left(\frac{\frac{1}{4}|z|^2}{|\nu_0+\frac{3}{2}|}-1\right),$$

where $|\nu_0+\frac{3}{2}|$ is the smallest of the numbers $|\nu+\frac{3}{2}|$, $|\nu+\frac{5}{2}|$, $|\nu+\frac{9}{2}|,\ldots$.

Correct

$$|\vartheta|<\frac{2}{3}\left(\exp\left(\frac{\frac{1}{4}|z|^2}{|\nu_0+\frac{3}{2}|}\right)-1\right).$$

Evidence $\nu=1000$, $z=\mathrm{i}y$ with $y^2/4=|\nu+\frac{3}{2}|$ gives $|\vartheta|>0.93$.

Lean `struveH_relative_bound` (`DLMF/Struve/Properties.lean`), admitted.

Found 2026-10-20 (`REVIEW.md`, "Project-wide admission triage and statement corrections").

a.15 (11.7.12)

Printed

$$\int_0^\infty t^{-\mu-\nu}\mathbf{H}_\mu(t)\,\mathbf{H}_\nu(t)\,\mathrm{d}t = \frac{\sqrt{\pi}\,\Gamma(\mu+\nu)}{2^{\mu+\nu}\Gamma\left(\mu+\nu+\frac{1}{2}\right)\Gamma\left(\mu+\frac{1}{2}\right)\Gamma\left(\nu+\frac{1}{2}\right)}, \qquad \Re(\mu+\nu)>0.$$

Correct the condition $\Re(\mu+\nu)>0$ alone is insufficient; the integral also needs $\Re\mu > -\frac{3}{2}$ and $\Re\nu > -\frac{3}{2}$ ($\mu=3$, $\nu=-2$ diverges).

Lean `struveH_integral_product` (`DLMF/Struve/Representations.lean`), admitted.

Found 2026-10-20 (as (11.4.15)).

a.16 (11.11.6)

Printed for fixed z and $\nu\to\infty$ in $|\operatorname{ph}\nu|\le\pi$, bounded away from the integers,

$$\mathbf{E}_\nu(z) = \frac{2}{\pi\nu}\left(\sin^2\left(\tfrac{1}{2}\pi\nu\right) + \frac{\nu z}{\nu^2-1}\cos^2\left(\tfrac{1}{2}\pi\nu\right) + O\left(\frac{1}{\nu^2}\right)\right).$$

Correct for complex ν the error term is $O(\mathrm{e}^{\pi|\Im\nu|}\nu^{-2})$ instead of $O(\nu^{-2})$.

Evidence at $z=1$, $|\text{error}|\cdot|\nu|^2 \approx 1.05\cdot 10^{13}$ at $\nu=\frac{1}{2}+10\mathrm{i}$.

Lean `weberE_large_order` (`DLMF/Struve/Asymptotics.lean`), proved.

Found 2026-10-21 (`REVIEW.md`, "Round 6 (2026-10-21)").

a.17 (11.11.18), (11.11.19)

Printed as $\nu\to\infty$ in $|\operatorname{ph}\nu|\le\pi-\delta$ (DLMF's 2021 clarification widens the condition to this sector),

$$\mathbf{J}_\nu(\nu) \sim \frac{2^{1/3}}{3^{2/3}\Gamma\left(\frac{2}{3}\right)\nu^{1/3}}, \qquad \mathbf{E}_\nu(\nu) \sim \frac{2^{1/3}}{3^{7/6}\Gamma\left(\frac{2}{3}\right)\nu^{1/3}}.$$

Correct the relations are correct only for real $\nu\to+\infty$.

Evidence the ratio is 1 for real ν but $\approx 2.8\cdot 10^6$ at $\operatorname{ph}\nu=0.3$ (matching $\sin(\pi\nu)\mathbf{A}_\nu(\nu)$).

Lean `angerJ_turning_equiv`, `weberE_turning_equiv` (`DLMF/Struve/Asymptotics.lean`), proved.

Found 2026-10-21 (as (11.11.6)).

2.5 Chapter 12 (parabolic cylinder functions)

a.18 (12.13.6)

Printed for $n=0,1,2,\ldots$,

$$n!\,U\left(n+\tfrac{1}{2},z\right) = \mathrm{i}^n\mathrm{e}^{-\frac{1}{2}z^2}\operatorname{erfc}(z/\sqrt{2})\,U\left(-n-\tfrac{1}{2},\mathrm{i}z\right) + \sum_{m=1}^{\lfloor\frac{1}{2}n+\frac{1}{2}\rfloor} U\left(2m-n-\tfrac{1}{2},z\right).$$

Correct the formula fails at $n=0$, $z=0$: the right side is 1 but $U(\frac{1}{2},0)=\sqrt{\pi/2}$.

Lean `pcfU_half_zero_ne_one`, `dlmf_12_13_6_counterexample` (`DLMF/ParabolicCylinder/SourceChecks.lean`), proved.

Found 2026-10-19 (`DLMF111214Acquisition.md`, "Formally verified source discrepancies").

2.6 Chapter 13 (confluent hypergeometric functions)

a.19 (13.9.6)

Printed the number of positive zeros of $M(a,b,x)$:

$$p(a,b) = \left\lfloor \tfrac{1}{2}\left(\lceil -b\rceil - \lceil -a\rceil + 1\right)\right\rfloor - \left\lfloor \tfrac{1}{2}\left(\lceil -b\rceil - \lceil -a\rceil\right)\right\rfloor, \qquad \lceil -b\rceil > \lceil -a\rceil > 0.$$

Correct the count needs $a \notin -\mathbb{N}$: $a=-1$, $b=-3/2$ gives $M = 1+2x/3$ with no positive zero, while the formula gives one.

Lean `kummerM_posZeros_ceil_swap`, counterexample `kummerM_posZeros_ceil_swap_ne` (`DLMF/Confluent/Zeros.lean`), proved.

Found 2026-10-21 (`REVIEW.md`, "Round 7 statement corrections").

a.20 (13.11.3) (borderline (a)/(c): branch)

Printed

$$\mathbf{M}\left(a,b,z\right) = \mathrm{e}^{\frac{1}{2}z}\sum_{s=0}^{\infty} A_s\left(b-2a\right)^{\frac{1}{2}(1-b-s)}\left(\tfrac{1}{2}z\right)^{\frac{1}{2}(1-b+s)} J_{b-1+s}\left(\sqrt{2z(b-2a)}\right).$$

Correct with principal branches the printed Bessel expansion is false when $\operatorname{ph} z + \operatorname{ph}(b-2a) > \pi$.

Evidence $a = 0.2-0.5\mathrm{i}$, $b = 1.4+0.4\mathrm{i}$, $z = -1.2+0.3\mathrm{i}$: the right side is $2.8336-3.3899\mathrm{i}$ against $1.18529+0.46781\mathrm{i}$.

Lean `regularizedHGFun_eq_besselExpansion` (branch-free, all a, b, z), `regularizedHGFun_besselJ_series` (printed form with the phase condition) (`DLMF/Confluent/Series.lean`), proved.

Found 2026-10-20 (`REVIEW.md`, "Blocker round (2026-10-20, late evening)").

2.7 Chapter 14 (Legendre and related functions)

a.21 (14.14.1)

Printed

$$\tfrac{1}{2}\left(x^2-1\right)^{1/2}\frac{P_\nu^{\mu}\left(x\right)}{P_\nu^{\mu-1}\left(x\right)} = \cfrac{x_0}{y_0+\cfrac{x_1}{y_1+\cfrac{x_2}{y_2+\cdots}}},$$

with $x_k = \frac{1}{4}(\nu-\mu-k+1)(\nu+\mu+k)(x^2-1)$, $y_k = (\mu+k)x$ (14.14.2).

Correct the continued fraction is the minimal-solution ratio

$$\tfrac{1}{2}\left(x^2-1\right)^{1/2}(\nu+\mu)(\nu-\mu+1)\frac{P_\nu^{-\mu}\left(x\right)}{P_\nu^{1-\mu}\left(x\right)} = \cfrac{x_0}{y_0+\cfrac{x_1}{y_1+\cfrac{x_2}{y_2+\cdots}}}.$$

Evidence $(\nu,\mu,x) = (0.3, 0.4, 1.5)$: the fraction is 0.345161187628545, the printed left side 0.812720708269563.

Lean `associatedP_continued_fraction` (`DLMF/Legendre/Series.lean`), proved.

Found 2026-10-21 (`REVIEW.md`, "Round 7 (2026-10-21)").

a.22 §14.16(ii), Ferrers P zeros

Printed the number of zeros of $\mathsf{P}^{\mu}_{\nu}(x)$ in $(-1,1)$ is $\max(\lceil \nu - |\mu| \rceil, 0) + 1$ if (a) $\mu > 0$, $n > m$, and $\delta_\nu \le \delta_\mu$, or (b) $\mu > 0$, $n < m$, and $m - n$ is even.

Correct the extra-zero case (a) needs $n \ge m$, not $n > m$ (the case $n = m$ is uncovered).

Evidence $(\nu, \mu) = (1/4, 3/4)$, $(0.3, 0.3)$, $(1.6, 1.6)$ each have one zero against the base count 0.

Lean `ferrersPExtraZero`, `ferrersP_zero_count` (`DLMF/Legendre/Zeros.lean`), proved.

Found 2026-10-22 (`REVIEW.md`, "Round 9 DLMF errata in §14.16 (2026-10-22, user-approved)").

a.23 §14.16(ii), Ferrers Q zeros

Printed $\mathsf{Q}^{\mu}_{\nu}(x)$ has $\max(\lceil \nu - |\mu| \rceil, 0) + k$ zeros in $(-1,1)$, $k \in \{-1, 0, 1, 2\}$, with $\max(\lceil \nu - |\mu| \rceil, 0) + k$ even or odd according as $\cos(\nu\pi)$ and $\cos(\mu\pi)$ have opposite signs or the same sign.

Correct the parity rule holds only for $\mu \ge 0$ ($\nu = -0.369$, $\mu = -0.206$: $\mathsf{Q} > 0$ on $(-1,1)$).

Lean `ferrersQ_zero_count_range` (`DLMF/Legendre/Zeros.lean`), proved.

Found 2026-10-22 (same section).

a.24 §14.16(iii), zeros of P on $(1,\infty)$

Printed $P^{\mu}_{\nu}(x)$ has exactly one zero in $(1,\infty)$ if (a) $\mu > 0$, $\mu > \nu$, $\mu \notin \mathbb{Z}$, and $\sin((\mu-\nu)\pi)$ and $\sin(\mu\pi)$ have opposite signs, or (b) $\mu \le \nu$, $\mu \notin \mathbb{Z}$, and $\lfloor \mu \rfloor$ is odd.

Correct case (b) needs $\mu > 0$ ($\nu = 0.184$, $\mu = -0.088$: $P > 0$ on $(1,\infty)$).

Lean `associatedPOneZero`, `associatedP_zero_count` (`DLMF/Legendre/Zeros.lean`), proved.

Found 2026-10-22 (same section).

a.25 (14.23.7)

Printed

$$\widehat{\mathsf{Q}}^{-\mu}_{-\frac{1}{2}+\mathrm{i}\tau}(x) = \tfrac{1}{2}\mathrm{e}^{3\mu\pi\mathrm{i}/2} Q^{-\mu}_{-\frac{1}{2}+\mathrm{i}\tau}(x - \mathrm{i}0) + \tfrac{1}{2}\mathrm{e}^{-3\mu\pi\mathrm{i}/2} Q^{-\mu}_{-\frac{1}{2}-\mathrm{i}\tau}(x + \mathrm{i}0)\,.$$

Correct the second factor is $\mathrm{e}^{\mu\pi\mathrm{i}/2}$ (the two terms are then conjugate):

$$\widehat{\mathsf{Q}}^{-\mu}_{-\frac{1}{2}+\mathrm{i}\tau}(x) = \tfrac{1}{2}\mathrm{e}^{3\mu\pi\mathrm{i}/2} Q^{-\mu}_{-\frac{1}{2}+\mathrm{i}\tau}(x - \mathrm{i}0) + \tfrac{1}{2}\mathrm{e}^{\mu\pi\mathrm{i}/2} Q^{-\mu}_{-\frac{1}{2}-\mathrm{i}\tau}(x + \mathrm{i}0)\,.$$

Lean `conicalQHat_boundary` (`DLMF/Legendre/Continuation.lean`), proved.

Found 2026-10-20 (`REVIEW.md`, "Shared-helper round").

a.26 §14.27, left-cut zero rule

Printed $P^{\mu}_{\nu}(x \pm \mathrm{i}0)$ has exactly one zero in $(-\infty,-1)$ if (a) $\mu < 0$, $\mu \notin \mathbb{Z}$, $\nu \in \mathbb{Z}$, and $\sin((\mu-\nu)\pi)$ and $\sin(\mu\pi)$ have opposite signs, or (b) $\mu, \nu \in \mathbb{Z}$, $\mu + \nu < 0$, and ν is odd.

Correct the rule predicts one zero for degree $\nu = 1$, order $\mu = -1/2$; there is none.

Lean `associatedP_one_neg_half`, `dlmf_14_27_no_boundary_zero` (`DLMF/Legendre/SourceChecks.lean`), proved.

Found 2026-10-19 (`DLMF111214Acquisition.md`, "Formally verified source discrepancies").

2.8 Chapters 16, 17 and 24

a.27 (16.19.5) (borderline (a)/(c))

Printed with $\vartheta = z\,\mathrm{d}/\mathrm{d}z$,

$$\vartheta\, G^{m,n}_{p,q}\left(z;\begin{matrix}a_1,\dots,a_p\\ b_1,\dots,b_q\end{matrix}\right) = G^{m,n}_{p,q}\left(z;\begin{matrix}a_1-1,a_2,\dots,a_p\\ b_1,\dots,b_q\end{matrix}\right) + (a_1-1)\,G^{m,n}_{p,q}\left(z;\begin{matrix}a_1,\dots,a_p\\ b_1,\dots,b_q\end{matrix}\right).$$

Correct this plus-sign transformation needs $n \geq 1$; for $n = 0$ the sign is opposite.

Lean `meijerG_log_derivative` (hypothesis `0 < n`) (`DLMF/GeneralizedHypergeometric/MeijerGRelations.lean`), admitted.

Found 2026-10-24 (`DLMF16Acquisition.md`, 16.19.E5 row).

a.28 (17.2.34)

Printed

$$\lim_{n\to\infty}\begin{bmatrix}rn+u\\ sn+t\end{bmatrix}_q = \frac{1}{(q;q)_\infty} = \prod_{j=1}^{\infty}\frac{1}{1-q^j}, \qquad r > s.$$

Correct the condition $r > s$ omits $s > 0$; at $s = 0$ the limit is $1/(q;q)_t$.

Lean `qBinom_two_growing_indices`, `qBinom_two_index_limit` (`DLMF/QHypergeometric/Calculus.lean`), proved.

Found 2026-10-24 (`DLMF17Acquisition.md`, 17.2.E34 row).

a.29 (17.7.19)

Printed

$${}_{r+1}\phi_r\left(\begin{matrix}a, b_1q^{m_1},\dots,b_rq^{m_r}\\ b_1,\dots,b_r\end{matrix};q,a^{-1}q^{1-(m_1+\cdots+m_r)}\right) = 0.$$

Correct the printed argument has an extra q; the argument is $a^{-1}q^{-M}$, $M = m_1+\cdots+m_r$ (Gasper–Rahman [6, (1.9.7)]):

$${}_{r+1}\phi_r\left(\begin{matrix}a, b_1q^{m_1},\dots,b_rq^{m_r}\\ b_1,\dots,b_r\end{matrix};q,a^{-1}q^{-(m_1+\cdots+m_r)}\right) = 0.$$

Evidence exact counterexample $r = 0$, $q = 1/2$, $a = 2$, $M = 0$: the printed value is $1/2$, not 0.

Lean `karlsson_printed_counter` (proved), `q_karlsson_minton_zero_corrected` (admitted) (`DLMF/QHypergeometric/HigherSummations.lean`).

Found 2026-10-24 (`DLMF17Acquisition.md`, "Domains and source corrections").

a.30 (17.7.23)

Printed

$$\left(1-\frac{a}{q}\right)\left(1-\frac{b}{q}\right)\sum_{k=0}^{n}\frac{(ap^k,bp^{-k};q)_{n-1}\,(1-ap^{2k}/b)}{(p;p)_n\,(p;p)_{n-k}\,(ap^k/b;q)_{n+1}}(-1)^kp^{\binom{k}{2}} = \delta_{n,0}.$$

Correct $(p;p)_k$ in place of $(p;p)_n$, and $(ap^k/b;p)_{n+1}$ in place of $(ap^k/b;q)_{n+1}$:

$$\left(1-\frac{a}{q}\right)\left(1-\frac{b}{q}\right)\sum_{k=0}^{n}\frac{(ap^k,bp^{-k};q)_{n-1}\,(1-ap^{2k}/b)}{(p;p)_k\,(p;p)_{n-k}\,(ap^k/b;p)_{n+1}}(-1)^kp^{\binom{k}{2}} = \delta_{n,0}.$$

Evidence the printed expression at $p = 1/2$, $q = 1/3$, $a = 1/4$, $b = 1$, $n = 1$ equals $-1884/1771$.

Lean `bibasic_printed_counter` (proved), `bibasic_difference_corrected` (admitted) (`DLMF/QHypergeometric/HigherSummations.lean`).

Found 2026-10-24 (same section).

a.31 (17.8.8)

Printed

$$ {}_2\psi_2\left(\begin{matrix} b^2, b^2/c \\ q, cq \end{matrix}; q^2, cq^2/b^2\right) = \frac{1}{2}\,\frac{(q^2, qb^2, q/b^2, cq/b^2; q^2)_\infty}{(cq, cq^2/b^2, q^2/b^2, c/b^2; q^2)_\infty} \times \left(\frac{(c\sqrt{q}/b; q)_\infty}{(b\sqrt{q}; q)_\infty} + \frac{(-c\sqrt{q}/b; q)_\infty}{(-b\sqrt{q}; q)_\infty}\right), \qquad |cq^2| < |b^2|. $$

Correct the printed condition $|cq^2| < |b^2|$ is broader than convergence; the sum identity needs $|c/b^2| < 1$.

Lean `bilateral_even_sum` (`DLMF/QHypergeometric/BilateralSummations.lean`), admitted.

Found 2026-10-24 (`DLMF17Acquisition.md`, 17.8.E8 row).

a.32 (17.14.3)

Printed the left side repeats that of (17.14.2):

$$ \sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(q^2;q^2)_n\,(-q;q^2)_{n+1}} = \text{coeff. of } z^0 \text{ in } \frac{(-zq;q^2)_\infty\,(-z^{-1}q;q^2)_\infty\,(q^2;q^2)_\infty}{(z^{-1};q^2)_\infty\,(-q;q^2)_\infty\,(z^{-1}q;q^2)_\infty} $$
$$ = \frac{1}{(-q;q^2)_\infty} \text{coeff. of } z^0 \text{ in } \frac{(-zq;q^2)_\infty\,(-z^{-1}q;q^2)_\infty\,(q^2;q^2)_\infty}{(z^{-1};q)_\infty} = \frac{G(q)}{(-q;q^2)_\infty}. $$

Correct $(-q;q^2)_n$ instead of $(-q;q^2)_{n+1}$ on the left side:

$$ \sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(q^2;q^2)_n\,(-q;q^2)_n} = \text{coeff. of } z^0 \text{ in } \cdots = \frac{G(q)}{(-q;q^2)_\infty}, $$

the constant-term expressions being unchanged.

Evidence coefficient comparison with the displayed constant term.

Lean `rogers_constantTerm_G_corrected` (`DLMF/QHypergeometric/ConstantTerms.lean`), admitted.

Found 2026-10-24 (same section as (17.7.19)).

a.33 (24.4.7) (borderline (a)/(c): 0^0 convention)

Printed

$$ \sum_{k=1}^{m} k^n = \frac{B_{n+1}\,(m+1) - B_{n+1}}{n+1}. $$

Correct the printed sum from $k = 1$ fails at $n = 0$; Mathlib's form, with the sum from $k = 0$, holds.

Lean upstream pointer to `Polynomial.sum_range_pow_eq_bernoulli_sub`.

Found 2026-10-21 (`REVIEW.md`, "Upstream pointers (2026-10-21)").

2.9 Chapter 19 (elliptic integrals)

Found by the Chapter 19 acquisition agents and checked by an independent mpmath verifier on 2026-10-25 against the printed formulas (DLMF 1.2.8); see `DLMF19Acquisition.md`.

a.34 (19.7.2) (second and fourth formulas)

Printed with no condition,

$$K\left(-\mathrm{i}k'/k\right) = kK\left(k'\right), \qquad E\left(-\mathrm{i}k'/k\right) = (1/k)E\left(k'\right).$$

The left sides depend on k^2 only, the right sides are odd in k. DLMF's condition $\Re k > 0$ (its erratum for V1.2.0) stands above (19.7.3) only.

Correct for $\Re k > 0$.

Evidence at $k = -0.6$, $K(-\mathrm{i}k'/k) = 1.19718$ but $kK(k') = -1.19718$; at $k = -0.6 + 0.3\mathrm{i}$ the two sides are $1.2490 - 0.3256\mathrm{i}$ and its negative.

Lean `compEllInt_imaginary_comodulus` (`DLMF/EllipticIntegrals/Connection.lean`), admitted.

Found 2026-10-25.

a.35 (19.19.3)

Printed

$$R_{-a}\left(\mathbf{b};\mathbf{z}\right) = z_n^{-a}\sum_{N=0}^{\infty}\frac{(a)_N}{(c)_N}\,T_N(b_1,\dots,b_{n-1};1-(z_1/z_n),\dots,1-(z_{n-1}/z_n)), \qquad |1-z_j/z_n|<1.$$

DLMF derives it from (19.16.11) and (19.19.2) but omits the condition of (19.16.11).

Correct with the additional condition

$$\operatorname{ph} z_n + \operatorname{ph}\left(z_j/z_n\right) \in (-\pi,\pi) \quad \text{for every } j.$$

Evidence $n = 2$, $a = 0.3$, $\mathbf{b} = (0.5, 0.7)$, $z_1 = \mathrm{e}^{-0.8\pi\mathrm{i}}$, $z_2 = \mathrm{e}^{0.9\pi\mathrm{i}}$ ($|1 - z_1/z_2| \approx 0.908$): $R = 1.8216 - 0.4034\mathrm{i}$ by the integral (19.16.9), against $0.5763 - 0.8336\mathrm{i}$ from the formula; the formula with the roles of z_1, z_2 exchanged gives a third value, $0.6147 + 0.8057\mathrm{i}$.

Lean `symEllIntR_hasSum_T_last` (`DLMF/EllipticIntegrals/SymmetricSeries.lean`), with the hypothesis `harg` added on 2026-10-25 (the acquired statement without it was false); proved.

Found 2026-10-25 (proof round 1).

a.36 (19.20.3), (19.20.23)

Printed under the §19.16(i) constraints,

$$R_F\left(x,a,y\right) = R_{-\frac14}\left(\tfrac34,\tfrac12;a^2,xy\right), \qquad R_D\left(x,y,a\right) = R_{-\frac34}\left(\tfrac54,\tfrac12;a^2,xy\right),$$
$$a = \tfrac12 x + \tfrac12 y.$$

They fail inside that domain.

Correct for $\Re x > 0$ and $\Re y > 0$ (sufficient).

Evidence at $x = -1 + \mathrm{i}$, $y = -1 + 2\mathrm{i}$ the ratio of right to left side is $+\mathrm{i}$ for (19.20.3) and $-\mathrm{i}$ for (19.20.23); at $x = -2 + 2\mathrm{i}$, $y = 1.5 - 1.5\mathrm{i}$ it is $0.945 + 0.328\mathrm{i}$ and $-0.541 + 0.841\mathrm{i}$; no failure at 85 random points with $\Re x, \Re y > 0$.

Lean `symEllIntRF_lemniscatic`, `symEllIntRD_lemniscatic` (real $x, y > 0$) and `symEllIntRF_lemniscatic_complex`, `symEllIntRD_lemniscatic_complex` ($\Re x, \Re y > 0$) (`DLMF/EllipticIntegrals/SymmetricSpecialCases.lean`), admitted. The acquisition first described the failure as a factor $-\mathrm{i}$ for all complex arguments; the verifier showed that the factor varies and that the formulas hold in the right half-planes, and the complex forms were added.

Found 2026-10-25.

a.37 §19.25(i): (19.25.5), (19.25.7), (19.25.9), (19.25.14)

Printed for complex ϕ in the strip $0 \le \Re\phi \le \pi/2$ (DLMF erratum V1.1.7) and complex k, with $c = \csc^2\phi$,

$$\begin{aligned} F(\phi,k) &= R_F(c-1, c-k^2, c), \\ E(\phi,k) &= 2R_G(c-1, c-k^2, c) - (c-1)R_F(c-1, c-k^2, c) \\ &\qquad - \sqrt{c-1}\,\sqrt{c-k^2}\big/\sqrt{c}, \\ E(\phi,k) &= R_F(c-1, c-k^2, c) - \tfrac{1}{3}k^2 R_D(c-1, c-k^2, c), \\ \Pi(\phi,\alpha^2,k) - F(\phi,k) &= \tfrac{1}{3}\alpha^2 R_J(c-1, c-k^2, c, c-\alpha^2). \end{aligned}$$

Correct With Legendre's integral along $[0,\phi]$ and principal branches these fail when ϕ and k^2 are both nonreal; they hold for real ϕ, and for complex ϕ with real k^2.

Evidence $\phi = 1.2 - 0.5\mathrm{i}$, $k^2 = 1.2 + 0.5\mathrm{i}$ gives $F = 1.0173 - 1.3254\mathrm{i}$ (continuous branch) or $1.0876 - 1.1700\mathrm{i}$ (principal) against $R_F = 2.1714 + 0.0724\mathrm{i}$.

Lean `incEllIntFk_eq_symEllIntRF`, `incEllIntEk_eq_symEllIntRG` and neighbours (`DLMF/EllipticIntegrals/SymmetricRelations.lean`), stated for real $\phi \in (0, \pi/2]$, admitted.

Found 2026-10-25.

a.38 (19.25.30), (19.25.31)

Printed

$$\begin{aligned} \operatorname{am}(u,k) &= R_C(\operatorname{cs}^2(u,k), \operatorname{ns}^2(u,k)) \quad \text{if } \operatorname{cs}^2(u,k) \ge 0, \\ u &= R_F(\operatorname{ps}^2(u,k), \operatorname{qs}^2(u,k), \operatorname{rs}^2(u,k)). \end{aligned}$$

The right sides are unchanged under $u \mapsto 2K - u$ and even in u.

Correct for $0 < u \le K$.

Evidence $k^2 = 0.5$, $u = 2.4 > K = 1.85407$: am $= 1.96619$ against $R_C = 1.17540$, and $R_F = 1.30815$ against u; at $u = -0.7$ the sign is wrong.

Lean `jacobiAmm_eq_symEllIntRC`, `eq_symEllIntRF_jacobi`, admitted.

Found 2026-10-25.

a.39 (19.25.32), (19.25.34)

Printed

$$\begin{aligned} \operatorname{arcps}(x,k) &= R_F(x^2, x^2 + \Delta(\mathrm{q},\mathrm{p}), x^2 + \Delta(\mathrm{r},\mathrm{p})), \\ \operatorname{arcpq}(x,k) &= \sqrt{w}\,R_F(x^2, 1, 1 + \Delta(\mathrm{r},\mathrm{q})w), \qquad w = (1-x^2)/\Delta(\mathrm{q},\mathrm{p}), \end{aligned}$$

with conditions on x^2 that allow negative x for arccs, arccn and arccd. The right sides are even in x and give $2K$ minus the true value for $x < 0$. Equation (19.25.33) is odd in x and correct as printed.

Correct for $x \ge 0$.

Evidence $k^2 = 0.5$: $\operatorname{arccs}(-0.6) = 2.58693$ against 1.12121; $\operatorname{arccn}(-0.6) = 2.71424$ against 0.99391; $\operatorname{arccd}(-0.6) = 2.51992$ against 1.18823.

Lean `jacobiArcps_eq_symEllIntRF`, `jacobiArcpq_eq_symEllIntRF`, admitted.

Found 2026-10-25.

a.40 (19.26.9)

Printed the R_J addition theorem

$$R_J(x+\lambda, y+\lambda, z+\lambda, p+\lambda) + R_J(x+\mu, y+\mu, z+\mu, p+\mu) \\ = R_J(x,y,z,p) - 3R_C(\gamma-\delta,\gamma),$$

with λ, x, y, z positive and no condition on p. It fails for some $p<0$.

Correct for $p>0$ (sufficient; it still holds for $-0.2004<p<0$ at the test point).

Evidence $(x,y,z,\lambda)=(1,2,4,0.5)$, $p=-3$: -0.132073 against -0.291838; $p=-1+2\mathrm{i}$: $0.15979-0.27854\mathrm{i}$ against $0.08663-0.46930\mathrm{i}$.

Lean `symEllIntRJ_add_symEllIntRJ`, admitted.

Found 2026-10-25.

a.41 (19.26.13)

Printed

$$R_C(\alpha^2, \alpha^2-\theta) + R_C(\beta^2, \beta^2-\theta) = R_C(\sigma^2, \sigma^2-\theta), \qquad \sigma = (\alpha\beta+\theta)/(\alpha+\beta),$$

with $0<\gamma^2-\theta<\gamma^2$ for $\gamma=\alpha,\beta,\sigma$; the signs of α,β are not fixed.

Correct for $\alpha,\beta>0$.

Evidence $\alpha=-2$, $\beta=3$, $\theta=1$ (all printed conditions hold): 0.895880 against 0.202733.

Lean `symEllIntRC_addition_theta`, admitted.

Found 2026-10-25.

a.42 (19.26.27)

Printed

$$R_C(x^2, x^2-\theta) = 2R_C(s^2, s^2-\theta), \qquad s = x+\sqrt{x^2-\theta}.$$

Correct The section's standing assumption $x>0$ is needed (a formalization note for complex x: $\Re x>0$), and, newly found by the verifier, $\theta<x^2$: for $\theta>x^2$ the left side is a Cauchy principal value, (19.2.20), but the right side is not. Correct for

$$\Re x>0 \quad \text{and} \quad x^2-\theta \in \mathbb{C}\setminus(-\infty,0].$$

Evidence $x=1$, $\theta=2$: 0.623225 against $0.623225-1.110721\mathrm{i}$ (difference $-\pi\mathrm{i}/(2\sqrt{\theta})$); $x=-2$, $\theta=-3$: 0.41207 against 1.40173; no failure at 3000 random complex points under the corrected hypotheses.

Lean `symEllIntRC_duplication_theta` (`DLMF/EllipticIntegrals/SymmetricAddition.lean`), admitted. The hypothesis $x^2-\theta\in\mathbb{C}\setminus(-\infty,0]$ was added on 2026-10-25: the acquired statement without it was false.

Found 2026-10-25.

a.43 (19.27.8)

Printed

$$R_D(x,y,z) = \frac{3}{\sqrt{xyz}} - \frac{6}{xy}R_G(x,y,0)\left(1+O\left(\frac{z}{g}\right)\right), \qquad z/g\to 0.$$

Correct the error term is $O(\sqrt{z/g})$:

$$R_D(x,y,z) = \frac{3}{\sqrt{xyz}} - \frac{6}{xy}R_G(x,y,0)\left(1+O\left(\sqrt{z/g}\right)\right).$$

Evidence $x = y = 1$, $z = 10^{-8}$: the relative error is $-1.273 \cdot 10^{-4}$; divided by z/g it grows without bound, divided by $\sqrt{z/g}$ it tends to -1.273 ($\approx -4/\pi$).

Lean `symEllIntRD_asymp_small_z` (`DLMF/EllipticIntegrals/SymmetricAsymptotics.lean`), admitted.

Found 2026-10-25.

a.44 (19.27.12)

Printed

$$R_J(x,y,z,p) = \frac{3}{2\sqrt{xyz}}\left(\ln\left(\frac{4f}{p}\right) - 2\right)\left(1 + O\left(\frac{p}{f}\right)\right), \qquad p/f \to 0.$$

The constant is right only for $x = y = z$, so the relative error is of order $1/\ln(1/p)$, not p/f.

Correct Replaced by Carlson and Gustafson (1994) [5, eq. (42)]:

$$R_J(x,y,z,p) = \frac{3}{2\sqrt{xyz}}\left(\ln\left(\frac{4f}{p}\right) - 2 + r\right),$$
$$-\ln(f/H) \le r \le \frac{3p}{2(f-p)}\ln(f/p), \qquad H = \frac{3}{x^{-1}+y^{-1}+z^{-1}}.$$

Evidence at $(1,2,4)$ the regularized constant tends to -0.015051, not $\ln(4f) - 2 = 0.079442$; the bounds on r hold at 500 random triples (checked numerically, not against the paper).

Lean `symEllIntRJ_small_p_bounds`, admitted.

Found 2026-10-25.

a.45 (19.28.9)

Printed with $x, y > 0$ only,

$$\int_0^{\pi/2} R_F\left(\sin^2\theta\cos^2(x+y), \sin^2\theta\cos^2(x-y), 1\right)\mathrm{d}\theta = R_F\left(0,\cos^2 x,1\right) R_F\left(0,\cos^2 y,1\right).$$

Correct for $x + y \le \pi/2$ (sufficient; the identity is also invariant under $x \mapsto \pi - x$).

Evidence $(x,y) = (1,1)$: 2.92136 against 4.35740; $(0.4, 1.4)$: 3.77223 against 5.19321.

Lean `integral_symEllIntRF_sin_cos` (`DLMF/EllipticIntegrals/SymmetricIntegralsOfIntegrals.lean`), admitted.

Found 2026-10-25.

2.10 Chapter 20 (theta functions)

a.46 (20.7.34)

Printed (in DLMF's nome notation, $q^2 \leftrightarrow 2\tau$, $\mathrm{i}q \leftrightarrow \tau + \frac{1}{2}$)

$$\frac{\theta_1(z,q^2)\,\theta_3(z,q^2)}{\theta_1(z,\mathrm{i}q)} = \frac{\theta_2(z,q^2)\,\theta_4(z,q^2)}{\theta_2(z,\mathrm{i}q)} = \mathrm{i}^{-1/4}\sqrt{\frac{\theta_2(0,q^2)\,\theta_4(0,q^2)}{2}},$$

that is,

$$\frac{\theta_1(z|2\tau)\,\theta_3(z|2\tau)}{\theta_1\left(z|\tau+\frac{1}{2}\right)} = \frac{\theta_2(z|2\tau)\,\theta_4(z|2\tau)}{\theta_2\left(z|\tau+\frac{1}{2}\right)} = \mathrm{i}^{-1/4}\left(\frac{\theta_2(0|2\tau)\,\theta_4(0|2\tau)}{2}\right)^{1/2}.$$

With the principal root this holds only up to sign.

Correct for all τ, the root on the right is

$$\mathrm{e}^{-\mathrm{i}\pi/8}\,\frac{\eta(\tau)\,\eta(4\tau)}{\eta(2\tau)} \qquad (\eta \text{ the Dedekind eta function}),$$

which equals the principal value for $\tau = \mathrm{i}t$.

Evidence the printed form has the opposite sign at $\tau = -2.19+1.29\mathrm{i}$, $2.03+0.90\mathrm{i}$, $-2.85+0.88\mathrm{i}$; the eta form agrees at 200 random (z,τ) (mpmath).

Lean `jacobiTheta_reduction_iq_sq` (squared form, all τ), `jacobiTheta_reduction_iq_rectangular` (printed form, $\tau = \mathrm{i}t$), `jacobiTheta_reduction_iq_eta` (full-domain form) (`DLMF/Theta/Identities.lean`), all admitted.

Found 2026-10-25.

a.47 (20.9.2)

Printed with no condition on τ,

$$K(k) = \tfrac{1}{2}\pi\,\theta_3^2(0|\tau), \qquad K'(k) = -\mathrm{i}\tau K(k).$$

With the principal K this fails for some complex τ.

Correct region: $K = \frac{1}{2}\pi\theta_3^2$ exactly on the translates $F + 2n$ of

$$F = \{\tau : \ |\Re\tau| < 1,\ |\tau - \tfrac{1}{2}| > \tfrac{1}{2},\ |\tau + \tfrac{1}{2}| > \tfrac{1}{2}\};$$

$K' = -\mathrm{i}\tau K$ on F.

Evidence 58 of 308 random τ fail; no failure at 676 points of F; the region matches the validity set at all of 500 random τ (mpmath).

Lean `compEllIntK_eq_jacobiThetaThreeNullTau` ($\tau = \mathrm{i}t$, real K), `compEllIntKk_eq_jacobiThetaThreeNullTau_fundamental` (complex principal K on F and $F + 2n$) (`DLMF/Theta/Relations.lean`), admitted.

Found 2026-10-25.

a.48 (20.9.3)

Printed without a branch condition,

$$R_F\left(\frac{\theta_2^2(z,q)}{\theta_2^2(0,q)}, \frac{\theta_3^2(z,q)}{\theta_3^2(0,q)}, \frac{\theta_4^2(z,q)}{\theta_4^2(0,q)}\right) = \frac{\theta_1'(0,q)}{\theta_1(z,q)}\,z.$$

The left side is π-periodic and even in real z, the right side $z\,\theta_1'(0)/\theta_1(z)$ is not; a condition keeping the arguments of R_F off its cut is needed.

Evidence fails at real $z = 1.6$, 2, 2.5 ($\tau = 0.8\mathrm{i}$); holds at $z = 1.4 + 0.5\mathrm{i}$, $\tau = 0.3 + 0.6\mathrm{i}$.

Correct Corrected complex forms (checked with mpmath): for $\tau = \mathrm{i}t$ the identity holds for complex $z \neq 0$ in the open period rectangle

$$|\Re z| < \pi/2, \qquad |\Im z| < \pi t/2$$

(no failure at 3500 points; fails at 373 of 500 points with $\pi t/2 < |\Im z| < 0.9\pi t$); for general τ it holds in a punctured neighbourhood of $z = 0$. For non-rectangular τ the period parallelogram $\pi(s + r\tau)$, $|s|, |r| < \frac{1}{2}$, is not a valid region (e.g. fails at $\tau = 0.448 + 0.520\mathrm{i}$, $s = -0.400$, $r = -0.318$).

Lean `carlsonRF_jacobiTheta` ($\tau = \mathrm{i}t$, real $0 < |z| \le \pi/2$), `carlsonRF_jacobiTheta_rect` and `carlsonRF_jacobiTheta_eventually` (`DLMF/Theta/Relations.lean`), all admitted, with the complex Chapter 19 R_F (`DLMF.EllipticIntegrals.symEllIntRF`).

Found 2026-10-25.

2.11 Chapter 21 (Riemann theta functions)

a.49 (21.5.4)

Printed for every symplectic $\mathbf{\Gamma} = \left[\begin{smallmatrix}\mathbf{A}&\mathbf{B}\\ \mathbf{C}&\mathbf{D}\end{smallmatrix}\right]$, with $\xi(\mathbf{\Gamma})^8 = 1$,

$$\theta\left(\left[[\mathbf{C\Omega}+\mathbf{D}]^{-1}\right]^{\mathrm{T}}\mathbf{z}\middle|[\mathbf{A\Omega}+\mathbf{B}][\mathbf{C\Omega}+\mathbf{D}]^{-1}\right) = \xi(\mathbf{\Gamma})\sqrt{\det[\mathbf{C\Omega}+\mathbf{D}]}\,\mathrm{e}^{\pi\mathrm{i}\mathbf{z}\cdot[[\mathbf{C\Omega}+\mathbf{D}]^{-1}\mathbf{C}]\cdot\mathbf{z}}\,\theta\left(\mathbf{z}|\mathbf{\Omega}\right).$$

False in general (it contradicts (21.5.7)).

Correct on the theta group ($\mathrm{diag}(\mathbf{CD}^{\mathrm{T}})$ and $\mathrm{diag}(\mathbf{AB}^{\mathrm{T}})$ even), with a continuous branch of $\sqrt{\det(\mathbf{C\Omega}+\mathbf{D})}$ and an eighth root of unity.

Evidence $g = 2$, $\mathbf{\Gamma} = \left[\begin{smallmatrix}\mathbf{I}&\mathbf{B}\\ \mathbf{0}&\mathbf{I}\end{smallmatrix}\right]$, $\mathbf{B} = \left[\begin{smallmatrix}1&3\\ 3&-1\end{smallmatrix}\right]$: the ratio $\theta(\mathbf{z}|\mathbf{\Omega}+\mathbf{B})/\theta(\mathbf{z}|\mathbf{\Omega})$ depends on $\mathbf{z}$.

Lean `riemannTheta_modular` (`DLMF/RiemannTheta/Modular.lean`), admitted.

Found 2026-10-25.

a.50 (21.5.8)

Printed

$$\mathbf{\Gamma} = \begin{bmatrix}\mathbf{0}_g & -\mathbf{I}_g\\ \mathbf{I}_g & \mathbf{0}_g\end{bmatrix} \;\Rightarrow\; \theta\left(\mathbf{\Omega}^{-1}\mathbf{z}\middle|-\mathbf{\Omega}^{-1}\right) = \sqrt{\det\left[-\mathrm{i}\mathbf{\Omega}\right]}\,\mathrm{e}^{\pi\mathrm{i}\mathbf{z}\cdot\mathbf{\Omega}^{-1}\cdot\mathbf{z}}\,\theta\left(\mathbf{z}|\mathbf{\Omega}\right),$$

with the principal square root of $\det(-\mathrm{i}\mathbf{\Omega})$, which is wrong for $g \geq 3$.

Correct the continuous branch of $\sqrt{\det(-\mathrm{i}\mathbf{\Omega})}$ with value 1 at $\mathbf{\Omega} = \mathrm{i}\mathbf{I}$ (the product of the principal roots of the eigenvalues).

Evidence $g = 3$, $\mathbf{\Omega} = \tau\mathbf{I}_3$, $\tau = -1.1\sqrt{3} + \mathrm{i}$: the ratio is minus the principal root.

Lean `riemannTheta_inversion` (continuous branch, admitted), `riemannTheta_inversion_principal` ($g \leq 2$, admitted), `riemannTheta_inversion_fin_one` ($g = 1$, proved) (`DLMF/RiemannTheta/Modular.lean`).

Found 2026-10-25.

a.51 (21.6.4)

Printed in

$$\prod_{j=1}^{h}\theta\begin{bmatrix}\sum_{k=1}^{h}T_{jk}\mathbf{c}_k\\ \sum_{k=1}^{h}T_{jk}\mathbf{d}_k\end{bmatrix}\left(\sum_{k=1}^{h}T_{jk}\mathbf{z}_k\middle|\mathbf{\Omega}\right) = \frac{1}{\mathcal{D}^g}\sum_{\mathbf{A}\in\mathcal{K}}\sum_{\mathbf{B}\in\mathcal{K}}\mathrm{e}^{-2\pi\mathrm{i}\sum_{j=1}^{h}\mathbf{b}_j\cdot\mathbf{c}_j}\prod_{j=1}^{h}\theta\begin{bmatrix}\mathbf{a}_j+\mathbf{c}_j\\ \mathbf{b}_j+\mathbf{d}_j\end{bmatrix}\left(\mathbf{z}_j|\mathbf{\Omega}\right),$$

"$\mathbf{c}_j$ and $\mathbf{d}_j$ are arbitrary h-dimensional vectors".

Correct $\mathbf{c}_j$, $\mathbf{d}_j$ are g-dimensional (characteristics of a genus-g theta function).

Evidence dimension count; independent numerical check of the identity.

Lean `riemann_identity_char` (`DLMF/RiemannTheta/Products.lean`), admitted.

Found 2026-10-25.

2.12 Chapter 22 (Jacobian elliptic functions)

a.52 Table 22.5.1

Printed the derivative of dc at $z = K + \mathrm{i}K'$ is k.

Correct $-k$ ($= k'^2$ sc nc there):

$$\frac{\mathrm{d}}{\mathrm{d}z}\,\mathrm{dc}\,(z,k)\Big|_{z=K+\mathrm{i}K'} = -k.$$

Evidence numerical differentiation.

Lean `jacobiDck_specialDerivs` (`DLMF/JacobiElliptic/SpecialValues.lean`), admitted.

Found 2026-10-25.

a.53 (22.8.23)

Printed for $z_1 + z_2 + z_3 + z_4 = 2K$,

$$\begin{vmatrix} \operatorname{sn} z_1 \operatorname{cn} z_1 & \operatorname{cn} z_1 \operatorname{dn} z_1 & \operatorname{cn} z_1 & \operatorname{dn} z_1 \\ \operatorname{sn} z_2 \operatorname{cn} z_2 & \operatorname{cn} z_2 \operatorname{dn} z_2 & \operatorname{cn} z_2 & \operatorname{dn} z_2 \\ \operatorname{sn} z_3 \operatorname{cn} z_3 & \operatorname{cn} z_3 \operatorname{dn} z_3 & \operatorname{cn} z_3 & \operatorname{dn} z_3 \\ \operatorname{sn} z_4 \operatorname{cn} z_4 & \operatorname{cn} z_4 \operatorname{dn} z_4 & \operatorname{cn} z_4 & \operatorname{dn} z_4 \end{vmatrix} = 0.$$

The printed determinant is nonzero.

Correct corrected forms: third column $\operatorname{sn} z_j$, or second column $\operatorname{sn} z_j \operatorname{dn} z_j$.

Evidence determinant $\approx -0.107 + 0.033\mathrm{i}$ at $k = 0.6$ ($\approx -0.03 + 0.04\mathrm{i}$ in the independent check).

Lean `jacobi_det_sum_two_K`, `jacobi_det_sum_two_K'` (`DLMF/JacobiElliptic/Addition.lean`), admitted.

Found 2026-10-25.

a.54 (22.9.23)

Printed the right side repeats $s_{2,3}d_{2,3}$:

$$s^{(4)}_{1,3}d^{(4)}_{1,3}c^{(4)}_{2,3}c^{(4)}_{3,3} + s^{(4)}_{2,3}d^{(4)}_{2,3}c^{(4)}_{3,3}c^{(4)}_{1,3} + s^{(4)}_{3,3}d^{(4)}_{3,3}c^{(4)}_{1,3}c^{(4)}_{2,3} = \frac{\kappa^2}{1-\kappa^2}\left(s^{(4)}_{1,3}d^{(4)}_{1,3} + s^{(4)}_{2,3}d^{(4)}_{2,3} + s^{(4)}_{2,3}d^{(4)}_{2,3}\right).$$

Correct last summand $s^{(4)}_{3,3}d^{(4)}_{3,3}$:

$$\cdots = \frac{\kappa^2}{1-\kappa^2}\left(s^{(4)}_{1,3}d^{(4)}_{1,3} + s^{(4)}_{2,3}d^{(4)}_{2,3} + s^{(4)}_{3,3}d^{(4)}_{3,3}\right).$$

Evidence $k = 0.6$, $z = 0.37 + 0.21\mathrm{i}$: printed right side $\approx 4.208 - 0.036\mathrm{i}$, left side $\approx 0.552 + 0.183\mathrm{i}$.

Lean `jacobi_cyclic_sdcc_three` (`DLMF/JacobiElliptic/Cyclic.lean`), admitted.

Found 2026-10-25.

2.13 Chapter 23 (Weierstrass elliptic and modular functions)

a.55 (23.6.17)

Printed with the principal $K(k)$,

$$K^2 = (K(k))^2 = \omega_1^2(e_1 - e_3), \qquad K'^2 = (K(k'))^2 = \omega_3^2(e_3 - e_1);$$

fails for some generator pairs.

Correct with the lattice $K = \frac{1}{2}\pi\,\theta_3^2(0|\tau)$ for every pair, or with the principal K when $\tau = \mathrm{i}K'(k)/K(k)$.

Evidence generators $2\omega_1 + 4\omega_3$, $2\omega_3$: $K^2 - \omega_1^2(e_1 - e_3) \approx 11.0 - 13.7\mathrm{i}$.

Lean `HalfPeriods.quarterK_sq_eq`, `HalfPeriods.compEllIntKk_sq_eq` (`DLMF/Weierstrass/Relations.lean`), admitted.

Found 2026-10-25.

a.56 (23.6.29)

Printed

$$\zeta\left(z|\mathbb{L}_3\right)-\zeta\left(z+2\mathrm{i}K'|\mathbb{L}_3\right)-\zeta\left(2\mathrm{i}K'|\mathbb{L}_3\right)=\mathrm{cs}\left(z,k\right).$$

Correct last term $+\zeta(2\mathrm{i}K'|\mathbb{L}_3)$:

$$\zeta\left(z|\mathbb{L}_3\right)-\zeta\left(z+2\mathrm{i}K'|\mathbb{L}_3\right)+\zeta\left(2\mathrm{i}K'|\mathbb{L}_3\right)=\mathrm{cs}\left(z,k\right).$$

Evidence the printed form is off by about $-0.49\mathrm{i}$ at $k^2=0.3$.

Lean `weierstrassZeta_cs` (`DLMF/Weierstrass/Relations.lean`), admitted.

Found 2026-10-25.

a.57 (23.8.1)

Printed for $q=\mathrm{e}^{\mathrm{i}\pi\omega_3/\omega_1}$, $\Im(z/\omega_1)<2\Im(\omega_3/\omega_1)$, $z\notin\mathbb{L}$,

$$\wp\left(z\right)+\frac{\eta_1}{\omega_1}-\frac{\pi^2}{4\omega_1^2}\csc^2\left(\frac{\pi z}{2\omega_1}\right)=-\frac{2\pi^2}{\omega_1^2}\sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}}\cos\left(\frac{n\pi z}{\omega_1}\right).$$

Correct condition $|\Im(z/\omega_1)|<2\Im(\omega_3/\omega_1)$.

Evidence outside the corrected region the series diverges (terms 10^5, 10^{11}, 10^{21} at $n=10$, 20, 40).

Lean `HalfPeriods.weierstrassP_fourier` (`DLMF/Weierstrass/Series.lean`), admitted.

Found 2026-10-25.

a.58 (23.8.2)

Printed under the same condition as (23.8.1),

$$\zeta\left(z\right)-\frac{\eta_1 z}{\omega_1}-\frac{\pi}{2\omega_1}\cot\left(\frac{\pi z}{2\omega_1}\right)=\frac{2\pi}{\omega_1}\sum_{n=1}^{\infty}\frac{q^{2n}}{1-q^{2n}}\sin\left(\frac{n\pi z}{\omega_1}\right).$$

Correct same condition correction as (23.8.1): $|\Im(z/\omega_1)|<2\Im(\omega_3/\omega_1)$.

Lean `HalfPeriods.weierstrassZeta_fourier` (`DLMF/Weierstrass/Series.lean`), admitted.

Found 2026-10-25.

a.59 (23.10.14)

Printed product over $\ell\neq j$:

$$A_n=n\prod_{j=0}^{n-1}\prod_{\substack{\ell=0\\ \ell\neq j}}^{n-1}\frac{1}{\sigma\left((2j\omega_1+2\ell\omega_3)/n\right)}.$$

Correct product over $(j,\ell)\neq(0,0)$:

$$A_n=n\prod_{\substack{0\le j,\ell\le n-1\\ (j,\ell)\neq(0,0)}}\frac{1}{\sigma\left((2j\omega_1+2\ell\omega_3)/n\right)}.$$

Evidence the printed index set gives relative error 0.71 at $n=2$.

Lean `HalfPeriods.nTupleConst` (`DLMF/Weierstrass/Addition.lean`), definition.

Found 2026-10-25.

a.60 (23.10.15)

Printed

$$A_n = \left(\frac{\pi^2 G^2}{\omega_1}\right)^{n^2-1} \frac{q^{n(n-1)/2}}{\mathrm{i}^{n-1}} \exp\left(-\frac{(n-1)\eta_1}{3\omega_1}\left((2n-1)(\omega_1^2+\omega_3^2)+3(n-1)\omega_1\omega_3\right)\right).$$

Correct π instead of π^2:

$$A_n = \left(\frac{\pi G^2}{\omega_1}\right)^{n^2-1} \frac{q^{n(n-1)/2}}{\mathrm{i}^{n-1}} \exp\left(-\frac{(n-1)\eta_1}{3\omega_1}\left((2n-1)(\omega_1^2+\omega_3^2)+3(n-1)\omega_1\omega_3\right)\right).$$

Evidence with π the two sides agree exactly; with π^2 they differ by the factor π^{n^2-1} ($n = 2, 3, 4$ on three lattices).

Lean `HalfPeriods.nTupleConst_eq` (`DLMF/Weierstrass/Addition.lean`), admitted.

Found 2026-10-25.

a.61 (23.11.2)

Printed with $\tau = \omega_3/\omega_1$ and f_1, f_2 as in (23.11.1),

$$\wp(z) = \frac{1}{z^2} + 8\int_0^\infty s\left(\mathrm{e}^{-s}\sinh^2\left(\tfrac{1}{2}zs\right) f_1(s,\tau) + \mathrm{e}^{\mathrm{i}\tau s}\sin^2\left(\tfrac{1}{2}zs\right) f_2(s,\tau)\right)\mathrm{d}s,$$

under the condition $-1 < \Re(z+\tau) < 1$, $|\Im z| < \Im\tau$.

Correct valid only for the normalization $2\omega_1 = 1$, and the printed condition $-1 < \Re(z+\tau) < 1$ does not give convergence. Corrected:

$$|\Re z| + |\Re\tau| < 1, \qquad |\Im z| < \Im\tau.$$

Evidence $\tau = -0.6 + \mathrm{i}$, $z = 0.6 + 0.1\mathrm{i}$: integrand 4.5, 68, 7452, $4.4 \cdot 10^7$ at $s = 10$, 20, 40, 80.

Lean `HalfPeriods.weierstrassP_integral` (`DLMF/Weierstrass/Integrals.lean`), admitted.

Found 2026-10-25.

a.62 (23.11.3)

Printed

$$\zeta(z) = \frac{1}{z} + \int_0^\infty \left(\mathrm{e}^{-s}\left(zs - \sinh(zs)\right) f_1(s,\tau) - \mathrm{e}^{\mathrm{i}\tau s}\left(zs - \sin(zs)\right) f_2(s,\tau)\right)\mathrm{d}s.$$

Correct the printed integral lacks the factor 4:

$$\zeta(z) = \frac{1}{z} + 4\int_0^\infty \left(\mathrm{e}^{-s}\left(zs - \sinh(zs)\right) f_1(s,\tau) - \mathrm{e}^{\mathrm{i}\tau s}\left(zs - \sin(zs)\right) f_2(s,\tau)\right)\mathrm{d}s;$$

normalization and domain as in (23.11.2).

Evidence the printed form is off by about 0.08 at $\tau = 0.3 + 0.9\mathrm{i}$, $z = 0.3 + 0.2\mathrm{i}$.

Lean `HalfPeriods.weierstrassZeta_integral` (`DLMF/Weierstrass/Integrals.lean`), admitted.

Found 2026-10-25.

a.63 (23.18.4) (terminology)

Printed the text calls J, which satisfies $J(\mathcal{A}\tau) = J(\tau)$, a modular form of level zero.

Correct J has a pole at $q = 0$ ((23.17.5)), so it is a modular function, not a modular form in the sense of §23.15(i).

Lean `kleinJ_smul`, `isModularFunction_kleinJ` (`DLMF/Weierstrass/ModularFunctions.lean`), admitted.

Found 2026-10-25.

2.14 Chapter 24 (Bernoulli and Euler polynomials)

a.64 (24.4.24)

Printed the multiplication formula with roots of unity, for $n = 1, 2, \dots$, $m = 2, 3, \dots$:

$$B_n(mx) = m^n B_n(x) + n \sum_{k=1}^{n} \sum_{j=0}^{k-1} (-1)^j \binom{n}{k} \left(\sum_{r=1}^{m-1} \frac{\mathrm{e}^{2\pi \mathrm{i}(k-j)r/m}}{(1 - \mathrm{e}^{2\pi \mathrm{i} r/m})^n} \right) (j + mx)^{n-1}.$$

It is false for every $n \geq 1$ (at $n = 1$ it is off by $-(m-1)$).

Correct derived by partial fractions, with $\zeta = \mathrm{e}^{2\pi \mathrm{i}/m}$:

$$B_n(mx) = m^n B_n(x) - n \sum_{k=1}^{n} \sum_{j=0}^{k-1} (-1)^j \binom{k-1}{j} \left(\sum_{r=1}^{m-1} \frac{\zeta^r}{(1-\zeta^r)^k} \right) (j + mx)^{n-1}.$$

Evidence $x = 0.37$, $m = 2$, $n = 1$: printed -0.76, while $B_1(0.74) = 0.24$; the corrected form agrees to about 10^{-24} for $n \leq 8$, $m \leq 6$ at real and complex x.

Lean `bernoulliPolyB_mul_eq_roots_of_unity` (`DLMF/BernoulliEuler/Properties.lean`), admitted.

Found 2026-10-25.

a.65 (24.4.8), (24.4.11) (missing conditions)

Printed

$$\sum_{k=1}^{m} (-1)^{m-k} k^n = \frac{E_n(m+1) + (-1)^m E_n(0)}{2},$$

$$\sum_{\substack{k=1 \\ (k,m)=1}}^{m} k^n = \frac{1}{n+1} \sum_{j=1}^{n+1} \binom{n+1}{j} \left(\prod_{p \mid m} (1 - p^{n-j}) B_{n+1-j} \right) m^j.$$

Correct DLMF (24.4.8) fails at $n = 0$ (as DLMF (24.4.7) does, see its entry under Chapters 16, 17 and 24; DLMF (24.4.26) received the condition $n > 0$ in erratum V1.0.5, but these two did not), so it needs $n \geq 1$. DLMF (24.4.11) fails at $m = 1$ for $n \geq 1$ (left side 1, right side 0), so it needs $m \geq 2$.

Lean `sum_Icc_neg_one_pow_mul_pow_eq_euler` ($n \geq 1$), `sum_coprime_pow` ($m \geq 2$) (`DLMF/BernoulliEuler/Properties.lean`), admitted.

Found 2026-10-25.

a.66 (24.10.3) (missing condition)

Printed Kummer's congruence

$$\frac{B_m}{m} \equiv \frac{B_n}{n} \pmod{p}, \qquad m \equiv n \not\equiv 0 \pmod{p-1}.$$

Correct it fails when $m = 1$ or $n = 1$; the condition $m, n \geq 2$ is needed.

Evidence $p = 3$, $m = 1$, $n = 3$ gives $B_1/1 = -1/2 \not\equiv 0 = B_3/3 \pmod 3$. A scan of $p < 30$, $m, n < 40$ finds 92 failures, all with $m = 1$ or $n = 1$.

Lean `kummer_congruence` ($m, n \geq 2$) (`DLMF/BernoulliEuler/Arithmetic.lean`), admitted.

Found 2026-10-25.

a.67 (24.12.11)

Printed for odd n, as $n \to \infty$ with $m\,(\geq 1)$ fixed,

$$y_{2m}^{(n)} \to m - \tfrac{1}{2}.$$

Correct $y_m^{(n)} \to m - \frac{1}{2}$ (the zeros of $\cos \pi x$). In fact $y_{2m}^{(n)}$ tends to $2m - \frac{1}{2}$, so the printed form fails for every $m \geq 1$, and already DLMF (24.12.9)–(24.12.10) place $y_2^{(n)}$ near $\frac{3}{2}$.

Evidence the zeros of E_n in $[\frac{1}{2}, \infty)$ for $n = 21, 41, 61$ are about 0.5, 1.50025, 2.49975, 3.50025, 4.49975, 5.50025 (sign changes, mpmath; found by p24b, recomputed independently).

Lean `r24b_tendsto_eulerRealZero_odd_counterexample` (proved), `tendsto_eulerRealZero_odd` (corrected statement, proved) (`DLMF/BernoulliEuler/Zeros.lean`).

Found 2026-10-25 (proof round 2).

a.68 (24.16.9) (branch, borderline (a)/(c))

Printed

$$\left(\frac{t}{\mathrm{e}^t - 1}\right)^x = \sum_{n=0}^{\infty} B_n^{(x)} \frac{t^n}{n!}, \qquad |t| < 2\pi.$$

Correct for complex t, $t/(\mathrm{e}^t - 1)$ is real and negative at $t \approx 2.0085 + 4.4072\mathrm{i}$ ($|t| \approx 4.84 < 2\pi$), so the principal power $(t/(\mathrm{e}^t - 1))^x$ is not the analytic branch on all of $|t| < 2\pi$. The formalization uses complex t with $|t| < \pi$, and real t with $|t| < 2\pi$.

Evidence $t = 1.9663 + 4.4315\mathrm{i}$, $x = \frac{1}{2}$: principal power $0.0032 + 0.8049\mathrm{i}$, series $-0.0032 - 0.8049\mathrm{i}$.

Lean `hasSum_norlundPolynomial` ($\|t\| < \pi$, complex), `hasSum_norlundPolynomial_real` ($|t| < 2\pi$, real) (`DLMF/BernoulliEuler/Generalizations.lean`), both proved (proof round 3, 2026-10-26).

Found 2026-10-25.

2.15 Chapter 25 (zeta and related functions)

a.69 (25.11.23)

Printed for $n = 1, 2, 3, \ldots$,

$$\zeta'\left(1 - 2n, \tfrac{1}{3}\right) = -\frac{\pi(9^n - 1)B_{2n}}{8n\sqrt{3}\,(3^{2n-1} - 1)} - \frac{B_{2n} \ln 3}{4n \cdot 3^{2n-1}} - \frac{(-1)^n \psi^{(2n-1)}\left(\frac{1}{3}\right)}{2\sqrt{3}\,(6\pi)^{2n-1}} - \frac{(3^{2n-1} - 1)\,\zeta'(1 - 2n)}{2 \cdot 3^{2n-1}}.$$

Correct the first denominator must be $8n\sqrt{3} \cdot 3^{2n-1}$, not $8n\sqrt{3}\,(3^{2n-1} - 1)$:

$$\zeta'\left(1 - 2n, \tfrac{1}{3}\right) = -\frac{\pi(9^n - 1)B_{2n}}{8n\sqrt{3} \cdot 3^{2n-1}} - \frac{B_{2n} \ln 3}{4n \cdot 3^{2n-1}} - \frac{(-1)^n \psi^{(2n-1)}\left(\frac{1}{3}\right)}{2\sqrt{3}\,(6\pi)^{2n-1}} - \frac{(3^{2n-1} - 1)\,\zeta'(1 - 2n)}{2 \cdot 3^{2n-1}}.$$

Evidence $n = 1$: $\zeta'(-1, \frac{1}{3}) = 0.0937262017607794$; printed $0.0433428\ldots$; the corrected form agrees to 15 digits (also $n = 2, 3, 4$) and with DLMF (25.11.21) at $h = 1$, $k = 3$.

Lean `deriv_hurwitzZeta_one_sub_two_mul_third` (`DLMF/Zeta/Hurwitz.lean`), admitted.

Found 2026-10-25.

a.70 (25.11.37) (branch)

Printed for $n = 2, 3, 4, \dots$, $\Re a \geq 1$,

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k} \zeta(nk, a) = -n \ln \Gamma(a) + \ln \left(\prod_{j=0}^{n-1} \Gamma\left(a - \mathrm{e}^{(2j+1)\pi \mathrm{i}/n}\right) \right).$$

Correct with principal logarithms the identity holds only modulo $2\pi\mathrm{i}$; it holds with the log-gamma function (that is, with $\ln \Gamma$ in place of the principal logarithms of $\Gamma(a)$ and of the product).

Evidence off by exactly $2\pi\mathrm{i}$ at $n = 2$, $a = 4 - 9\mathrm{i}$, and at $n = 3$, $a = 1 + 5\mathrm{i}$.

Lean `hasSum_hurwitzZeta_log_gamma` (real $a \geq 1$), `exp_tsum_hurwitzZeta_eq_prod_gamma` (exponentiated, $\Re a \geq 1$) (`DLMF/Zeta/HurwitzIntegrals.lean`), admitted.

Found 2026-10-25.

a.71 (25.12.13)

Printed for $z \in \mathbb{C} \setminus [0, \infty)$,

$$\mathrm{Li}_s(z) + \mathrm{e}^{\pi \mathrm{i} s} \mathrm{Li}_s\left(\frac{1}{z}\right) = \frac{(2\pi)^s \mathrm{e}^{\pi \mathrm{i} s/2}}{\Gamma(s)} \zeta\left(1 - s, \frac{\ln z}{2\pi \mathrm{i}}\right).$$

Correct with the principal $\ln z$ the formula fails for $\Im z < 0$ (and contradicts DLMF (25.12.4) at $s = 2$, $z = -2 - \mathrm{i}$); it holds with $0 < \operatorname{ph} z < 2\pi$, i.e. with parameter $\frac{1}{2} + \ln(-z)/(2\pi\mathrm{i})$:

$$\mathrm{Li}_s(z) + \mathrm{e}^{\pi \mathrm{i} s} \mathrm{Li}_s\left(\frac{1}{z}\right) = \frac{(2\pi)^s \mathrm{e}^{\pi \mathrm{i} s/2}}{\Gamma(s)} \zeta\left(1 - s, \frac{1}{2} + \frac{\ln(-z)}{2\pi \mathrm{i}}\right).$$

The error was introduced by DLMF's own V1.2.3 modification ($\mathrm{e}^{2\pi \mathrm{i} a}$ replaced by z).

Evidence $s = 2.5 + 0.2\mathrm{i}$, $z = 0.5 - 0.9\mathrm{i}$: left side $-0.19501 - 0.80944\mathrm{i}$, printed right side $4.3896 - 6.2751\mathrm{i}$.

Lean `polylog_add_exp_mul_polylog_inv` (`DLMF/Zeta/Polylog.lean`), admitted.

Found 2026-10-25.

a.72 (25.14.7) (missing condition)

Printed with no conditions,

$$\Phi(z, s, a) = \frac{\mathrm{i}(2\pi)^{s-1}}{z^a} \Gamma(1 - s) \Bigg(\mathrm{e}^{-\pi \mathrm{i} s/2} \Phi\left(\mathrm{e}^{-2\pi \mathrm{i} a}, 1 - s, \frac{\ln z}{2\pi \mathrm{i}}\right) - \mathrm{e}^{\pi \mathrm{i}(2a + (s/2))} \Phi\left(\mathrm{e}^{2\pi \mathrm{i} a}, 1 - s, 1 - \frac{\ln z}{2\pi \mathrm{i}}\right) \Bigg).$$

Correct the formula fails for $a = 1.6 + 0.2\mathrm{i}$, $a = 1.5$, $a = 1.2$, $a = -0.3 + 0.4\mathrm{i}$, and holds for about $0 < \Re a \leq 1$, $z \neq 0, 1$ (verified also at $|z| = 1$ and $z = 1.5 + 0.5\mathrm{i}$).

Lean `lerchPhi_transformation` (`DLMF/Zeta/Lerch.lean`), stated on the narrower verified domain $0 < \Re a < 1$, $0 < |z| < 1$, admitted.

Found 2026-10-25.

a.73 §25.15(ii) (prose)

Printed "There are also infinitely many zeros in the critical strip $0 \leq \Re s \leq 1$, located symmetrically about the critical line $\Re s = \frac{1}{2}$."

Correct for an even primitive χ the trivial zero $s = 0$ lies in that strip but $L(1, \chi) \neq 0$. The symmetry $\rho \mapsto 1 - \bar{\rho}$ holds for zeros $\rho \neq 0$.

Lean `q25b_LFunction_one_sub_conj_counterexample` (proved), `LFunction_one_sub_conj_eq_zero` (hypothesis $\rho \neq 0$ added in proof round 1) (`DLMF/Zeta/DirichletL.lean`), admitted.

Found 2026-10-25 (proof round 1).

2.16 Chapter 26 (combinatorial analysis)

a.74 (26.6.12) (missing condition)

Printed without a condition,

$$C(n) = \sum_{k=1}^{n} N(n,k).$$

Correct at $n = 0$ the sum is empty while $C(0) = 1$. The identity holds for $n \geq 1$.

Lean `q26a_not_catalan_eq_sum_narayana` (proved counterexample), `catalan_eq_sum_narayana` ($n \geq 1$, restated in proof round 1) (`DLMF/Combinatorics/LatticePaths.lean`), proved.

Found 2026-10-25 (proof round 1).

a.75 (26.8.31)

Printed "when $f(x)$ is analytic for all x, and the series converges",

$$\frac{1}{k!}\frac{\mathrm{d}^k}{\mathrm{d}x^k} f(x) = \sum_{n=k}^{\infty} \frac{s(n,k)}{n!}\,\Delta^n f(x).$$

Correct this hypothesis is insufficient: $f(z) = \sin(\pi z)$, $x = 0$, $k = 1$ has every $\Delta^n f(0) = 0$ but $f'(0) = \pi$. A Newton-series hypothesis is needed; the formula is acquired for polynomials.

Lean `not_forall_stirling_deriv_eq_fwdDiff_series` (proved), `hasSum_signedStirlingFirst_fwdDiff` (polynomials, admitted) (`DLMF/Combinatorics/Stirling.lean`).

Found 2026-10-25.

a.76 (26.9.7)

Printed "when $|xq| < 1$",

$$\sum_{m,n=0}^{\infty} p_k(\leq m, n)\, x^k q^n = 1 + \sum_{k=1}^{\infty} \begin{bmatrix} m+k \\ k \end{bmatrix}_q x^k = \prod_{j=0}^{m} \frac{1}{1 - x\,q^j}.$$

Correct the sum must run over k, n (printed m, n, with m free on the right), and the condition must be $|x| < 1$ (the printed $|xq| < 1$ is insufficient: $q = 0.5$, $x = 1.5$):

$$\sum_{k,n=0}^{\infty} p_k(\leq m, n)\, x^k q^n = 1 + \sum_{k=1}^{\infty} \begin{bmatrix} m+k \\ k \end{bmatrix}_q x^k = \prod_{j=0}^{m} \frac{1}{1 - x\,q^j}, \qquad |x| < 1.$$

Evidence power-series check for $m \leq 3$, $k < 6$.

Lean `hasSum_qBinom_mul_pow_x` (`DLMF/Combinatorics/RestrictedPartitions.lean`), admitted.

Found 2026-10-25.

a.77 (26.10.9)

Printed

$$p_m(\mathcal{D}, n) = p_m(\mathcal{D}, n-m) + p_{m-1}(\mathcal{D}, n).$$

This is false ($m = n = 2$: $1 \neq 2$; 43 failures for $m \leq 6$, $n < 15$).

Correct

$$p_m(\mathcal{D}, n) = p_{m-1}(\mathcal{D}, n) + p_m(\mathcal{D}, n-m) - p_{m-2}(\mathcal{D}, n-m),$$

equivalently $q_m(n) = q_m(n-m) + q_{m-1}(n-m)$ for partitions into exactly m distinct parts.

Lean `partitionCountAtMostDistinct_recurrence`, `card_distinct_card_eq_recurrence` (`DLMF/Combinatorics/PartitionIdentities.lean`), admitted.

Found 2026-10-25.

a.78 §26.10(iv) (prose)

Printed "$p(\mathcal{D}'3, n) \le p(\mathcal{D}3, n)$, with strict inequality for $n \ge 9$."

Correct the inequality is strict at $n = 9$ and for $n \ge 15$ (checked to $n = 30$) but an equality for $10 \le n \le 14$.

Lean `partitionCountSchur_le`, `partitionCountSchur_lt_iff` (`DLMF/Combinatorics/PartitionIdentities.lean`).

Found 2026-10-25.

a.79 (26.10.18)

Printed

$$A_k(n) = \sum_{\substack{1<h\le k\\(h,k)=1}} \mathrm{e}^{\pi \mathrm{i} f(h,k) - (2\pi i n h/k)}.$$

Correct the range must be $1 \le h \le k$ (the printed $1 < h \le k$ gives $A_1 = 0$):

$$A_k(n) = \sum_{\substack{1\le h\le k\\(h,k)=1}} \mathrm{e}^{\pi \mathrm{i} f(h,k) - (2\pi i n h/k)}.$$

Evidence with the correction, DLMF (26.10.17) gives 295.998 at $n = 30$ ($p(\mathcal{D}, 30) = 296$); with the printed range it gives $0.23 - 0.07\mathrm{i}$.

Lean `distinctPartitionKloosterman` (`DLMF/Combinatorics/PartitionIdentities.lean`), definition.

Found 2026-10-25.

a.80 (26.12.24)

Printed

$$\sum_{\substack{\pi\subseteq B(r,r,r)\\ \pi \text{ descending plane partition}}} q^{|\pi|} = \prod_{1\le h<j\le r} \frac{1-q^{r+h+j-1}}{1-q^{2h+j-1}}.$$

Correct the product must run over $1 \le h \le j \le r$:

$$\sum_{\substack{\pi\subseteq B(r,r,r)\\ \pi \text{ descending plane partition}}} q^{|\pi|} = \prod_{1\le h\le j\le r} \frac{1-q^{r+h+j-1}}{1-q^{2h+j-1}}.$$

Evidence at $r = 2$ the printed form gives $(1+q)(1+q^2)/(1+q+q^2)$ instead of $1+q^2$. Brute-force generating functions agree with the corrected form for $r = 1, \dots, 4$.

Lean `sum_descendingPPIn_pow` (`DLMF/Combinatorics/PlanePartitions.lean`), admitted.

Found 2026-10-25.

a.81 (26.12.12) (prose definition)

Printed "A plane partition is transpose complement if it is equal to the reflection through the (x, y)-plane of its complement. The number of transpose complement plane partitions in $B(r, r, 2t)$ is

$$\binom{t+r-1}{r-1} \prod_{1\le h\le j\le r-2} \frac{h+j+2t+1}{h+j+1}."$$

Correct read literally as reflection through the (x, y)-plane, the definition gives counts $0, 0, 2, 2, \dots$; the formula matches only with reflection in the plane $x = y$ (h and j swapped): $1, 1, 2, 3, 5, 14, 14$.

Lean `card_transposeComplement` (`DLMF/Combinatorics/PlanePartitions.lean`), admitted.

Found 2026-10-25.

a.82 (26.13.4), (26.7.3), (26.7.4), (26.4.10), (26.8.10), (26.8.29), (26.5.1) (missing conditions at $n = 0$ or $m = 0$)

Printed

$$d(n) = n!\sum_{j=0}^{n}(-1)^j\frac{1}{j!} = \left\lfloor \frac{n! + \mathrm{e} - 2}{\mathrm{e}} \right\rfloor, \tag{26.13.4}$$

$$B(n) = \sum_{k=1}^{m}\frac{k^n}{k!}\sum_{j=0}^{m-k}\frac{(-1)^j}{j!}, \quad m \geq n, \tag{26.7.3}$$

$$B(n) = \mathrm{e}^{-1}\sum_{k=1}^{\infty}\frac{k^n}{k!} = 1 + \left\lfloor \mathrm{e}^{-1}\sum_{k=1}^{2n}\frac{k^n}{k!} \right\rfloor, \tag{26.7.4}$$

$$\binom{n_1 + \cdots + n_m}{n_1, \ldots, n_m} = \sum_{k=1}^{m}\binom{n_1 + \cdots + n_m - 1}{n_1, \ldots, n_{k-1}, n_k - 1, n_{k+1}, \ldots, n_m}, \tag{26.4.10}$$

$$\sum_{k=1}^{n} S(n,k)\,(x-k+1)_k = x^n, \tag{26.8.10}$$

$$\sum_{k=1}^{n}(-1)^{n-k}s(n,k) = n!, \tag{26.8.29}$$

$$C(n) = \frac{1}{n+1}\binom{2n}{n} = \frac{1}{2n+1}\binom{2n+1}{n} = \binom{2n}{n} - \binom{2n}{n-1} = \binom{2n-1}{n} - \binom{2n-1}{n+1}. \tag{26.5.1}$$

In (26.4.10) DLMF states $n_1, \ldots, n_m \geq 1$.

Correct the floor form of (26.13.4) gives 0 at $n = 0$ ($d(0) = 1$); (26.7.3) needs $n \geq 1$ (at $n = 0$, $m = 2$ it gives $\frac{1}{2}$); the first (Dobiński) form of (26.7.4) gives $1 - \mathrm{e}^{-1}$ at $n = 0$; (26.4.10) fails at $m = 0$; (26.8.10) (sum from $k = 1$) and (26.8.29) fail at $n = 0$; the fourth form of (26.5.1) fails at $n = 0$.

Lean `numDerangements_eq_floor` (`DLMF/Combinatorics/Permutations.lean`), `bell_eq_sum_mul_sum`, `bell_eq_exp_neg_one_mul_tsum` (`Bell.lean`), `multinomial_eq_sum_update` (`Multinomial.lean`), `sum_Icc_stirlingSecond_mul_ascPochhammer`, `sum_neg_one_pow_mul_signedStirlingFirst` (`Stirling.lean`), `catalan_eq_choose_sub` (`Catalan.lean`); proved or admitted as the ledger says.

Found 2026-10-25.

a.83 Table 26.17.1 (missing conditions)

Printed among the entries of the twelvefold way:

$$S(n,1) + S(n,2) + \cdots + S(n,k), \qquad p_k(n) - p_{k-1}(n), \qquad \binom{k+n-1}{n}, \quad \binom{k}{n}, \quad \binom{n-1}{n-k}.$$

Correct the entry $S(n,1) + \cdots + S(n,k)$ fails at $n = 0$ (count 1, table 0; the sum should start at $j = 0$); $p_k(n) - p_{k-1}(n)$ is undefined at $k = 0$. The binomial entries need $\binom{-1}{0} = 1$.

Lean `twelvefoldCount_*` (`DLMF/Combinatorics/TwelvefoldWay.lean`), with side conditions.

Found 2026-10-25.

a.84 (26.18.4)

Printed the number of ways of placing n labeled objects into k labeled boxes so that at least one object is in each box is

$$k^n + \sum_{t=1}^{n}(-1)^t\binom{k}{t}(k-t)^n.$$

Correct the upper limit must be k:

$$k^n + \sum_{t=1}^{k} (-1)^t \binom{k}{t} (k-t)^n.$$

The printed form gives -3 at $n = 1$, $k = 3$, gives 4 at $n = 2$, $k = 4$, and gives 1 instead of 0 at $n = 0$, $k \geq 1$. The corrected form holds for all n, k ($0^0 = 1$).

Lean `card_surjective_eq_sum` (`DLMF/Combinatorics/CountingTechniques.lean`), admitted.

Found 2026-10-25.

2.17 Chapter 27 (functions of number theory)

a.85 (27.12.3)

Printed Legendre's formula, $x \geq 1$:

$$\pi(x) = \lfloor x \rfloor - 1 - \sum_{p_j \leq \sqrt{x}} \left\lfloor \frac{x}{p_j} \right\rfloor + \sum_{r \geq 2} (-1)^r \sum_{p_{j_1} < p_{j_2} < \cdots < p_{j_r} \leq \sqrt{x}} \left\lfloor \frac{x}{p_{j_1} p_{j_2} \cdots p_{j_r}} \right\rfloor.$$

As printed it computes $\pi(x) - \pi(\sqrt{x})$.

Correct

$$\pi(x) = \pi(\sqrt{x}) - 1 + \lfloor x \rfloor - \sum_{p_j \leq \sqrt{x}} \left\lfloor \frac{x}{p_j} \right\rfloor + \sum_{r \geq 2} (-1)^r \sum_{p_{j_1} < \cdots < p_{j_r} \leq \sqrt{x}} \left\lfloor \frac{x}{p_{j_1} \cdots p_{j_r}} \right\rfloor.$$

Evidence $x = 10$: printed 2, $\pi(10) = 4$, $\pi(\sqrt{10}) = 2$; the corrected form is checked for x up to 1000.

Lean `primePi_legendre` (`DLMF/NumberTheory/Primes.lean`), admitted.

Found 2026-10-25.

a.86 §27.3, first sentence

Printed "Except for $\nu(n)$, $\Lambda(n)$, p_n, and $\pi(x)$, the functions in §27.2 are multiplicative."

Correct false for ϕ_k, $k \geq 1$ (the sum of the k-th powers of the $m \leq n$ prime to n): $\phi_1(2)\phi_1(3) = 3 \neq \phi_1(6) = 6$, $\phi_2(2)\phi_2(3) = 5 \neq \phi_2(6) = 26$. The other §27.2 functions are multiplicative as claimed.

Lean `totientPowerSum_one_not_multiplicative` (`DLMF/NumberTheory/Functions.lean`), proved.

Found 2026-10-25.

a.87 (27.3.6) (missing condition)

Printed

$$\sigma_\alpha(n) = \prod_{r=1}^{\nu(n)} \frac{p_r^{\alpha(1+a_r)} - 1}{p_r^\alpha - 1}, \qquad \alpha \neq 0.$$

Correct for complex α, $\alpha \neq 0$ is insufficient: with $\alpha = 2\pi\mathrm{i}/\ln 2$, $2^\alpha = 1$ and the $p = 2$ factor is $0/0$ ($\sigma_\alpha(8) = 4$). The condition is $p_r^\alpha \neq 1$ for every prime p_r dividing n.

Lean `sigmaC_eq_prod` (`DLMF/NumberTheory/Functions.lean`), proved.

Found 2026-10-25.

2.18 Chapters 28–31 (Mathieu, Lamé, spheroidal and Heun functions)

All entries below were found by the acquisition agents on 2026-10-26 and re-derived by an independent verifier from the DLMF 1.2.8 page text with its own numerics; none is on the DLMF errata page.

a.88 (28.5.6), second relation

Printed

$$C_{2m}(-q) = C_{2m}(q), \qquad C_{2m+1}(-q) = S_{2m+1}(q), \qquad S_{2m+2}(-q) = S_{2m+2}(q).$$

Correct under DLMF's own sign rule (DLMF (28.5.5)–(28.5.7)) the second relation must read

$$C_{2m+1}(-q) = -S_{2m+1}(q).$$

Evidence by DLMF (28.2.35) and uniqueness, $f_{2m+1}(z,-q) = (-1)^{m+1} g_{2m+1}(\frac{1}{2}\pi - z, q)$; numerically $C_1(-0.5) = -0.494187$, $S_1(0.5) = +0.494187$ (also $n = 3$, $q = 0.2$, 1.0).

Lean `mathieuFeC_even_neg` and its neighbours (`DLMF/Mathieu/SecondSolutions.lean`), admitted.

a.89 (28.23.2)–(28.23.5)

Printed

$$\mathrm{me}_\nu(0,h^2)\,\mathrm{M}^{(j)}_\nu(z,h) = \sum_{n=-\infty}^{\infty} (-1)^n c^\nu_{2n}(h^2)\,\mathcal{C}^{(j)}_{\nu+2n}(2h\cosh z),$$

$$\mathrm{me}'_\nu(0,h^2)\,\mathrm{M}^{(j)}_\nu(z,h) = \mathrm{i}\tanh z \sum_{n=-\infty}^{\infty} (-1)^n (\nu+2n) c^\nu_{2n}(h^2)\,\mathcal{C}^{(j)}_{\nu+2n}(2h\cosh z),$$

"valid for all z when $j = 1$, and for $\Re z > 0$ and $|\cosh z| > 1$ when $j = 2, 3, 4$", and

$$\mathrm{me}_\nu(\tfrac{1}{2}\pi,h^2)\,\mathrm{M}^{(j)}_\nu(z,h) = \mathrm{e}^{\mathrm{i}\nu\pi/2} \sum_{n=-\infty}^{\infty} c^\nu_{2n}(h^2)\,\mathcal{C}^{(j)}_{\nu+2n}(2h\sinh z),$$

$$\mathrm{me}'_\nu(\tfrac{1}{2}\pi,h^2)\,\mathrm{M}^{(j)}_\nu(z,h) = \mathrm{i}\mathrm{e}^{\mathrm{i}\nu\pi/2}\coth z \sum_{n=-\infty}^{\infty} (\nu+2n) c^\nu_{2n}(h^2)\,\mathcal{C}^{(j)}_{\nu+2n}(2h\sinh z),$$

"valid for all z when $j = 1$, and for $\Re z > 0$ and $|\sinh z| > 1$ when $j = 2, 3, 4$".

Correct "valid for all z when $j = 1$" holds only for integer ν. For $\nu \notin \mathbb{Z}$ the terms with $n \to -\infty$ behave like $|\cosh z|^{-2|n|}$ ($|\sinh z|^{-2|n|}$ in (28.23.4)–(28.23.5)), so the series need the condition $|\cosh z| > 1$ (resp. $|\sinh z| > 1$).

Evidence $\nu = 0.5$, $h = 1$, $z = 0.5\mathrm{i}$: the terms at $n = -5, -20, -40$ are 0.027, 0.15, 9.8.

Lean `hasSum_modMathieuM_cosh`, `hasSum_modMathieuM_cosh_deriv`, `hasSum_modMathieuM_sinh`, `hasSum_modMathieuM_sinh_deriv` (`DLMF/Mathieu/Modified/BesselSeries.lean`), admitted with the condition $|\cosh z| > 1$ (resp. $|\sinh z| > 1$) for every j.

a.90 (28.28.11)

Printed

$$\int_0^\infty \mathrm{e}^{2\mathrm{i}h\cosh z\cosh t}\,\mathrm{Ce}_\nu(t,h^2)\,\mathrm{d}t = \tfrac{1}{2}\pi\mathrm{i}\,\mathrm{e}^{\mathrm{i}\nu\pi}\,\mathrm{ce}_\nu(0,h^2)\,\mathrm{M}^{(3)}_\nu(z,h).$$

Correct the factor $\mathrm{e}^{\mathrm{i}\nu\pi}$ must be $\mathrm{e}^{\mathrm{i}\nu\pi/2}$:

$$\int_0^\infty \mathrm{e}^{2\mathrm{i}h\cosh z\cosh t}\,\mathrm{Ce}_\nu(t,h^2)\,\mathrm{d}t = \tfrac{1}{2}\pi\mathrm{i}\,\mathrm{e}^{\mathrm{i}\nu\pi/2}\,\mathrm{ce}_\nu(0,h^2)\,\mathrm{M}^{(3)}_\nu(z,h).$$

Evidence term-by-term integration with DLMF (10.9.10) and DLMF (28.23.2) ($j=3$); numerically at $\nu=0.4$, $h=0.8$, $z=1+0.5\mathrm{i}$ the left side is $0.13373867-0.12414858\mathrm{i}$, the corrected right side $0.13373835-0.12414756\mathrm{i}$, the printed one $0.18117-0.02183\mathrm{i}$.

Lean `integral_exp_cosh_modMathieuCe` (`DLMF/Mathieu/Modified/Integrals.lean`), admitted.

a.91 (28.28.16)

Printed with the upper or lower signs according as $0\le y\le\pi$ or $\pi\le y\le 2\pi$,

$$\int_0^\infty \sin(2h\cos y\cosh t)\,\mathrm{Ce}_{2n}(t,h^2)\,\mathrm{d}t = -\frac{\pi A_0^{2n}(h^2)}{2\,\mathrm{ce}_{2n}(\frac12\pi,h^2)}\left(\mathrm{ce}_{2n}(y,h^2)\mp\frac{2}{\pi C_{2n}(h^2)}\,\mathrm{fe}_{2n}(y,h^2)\right).$$

Correct the overall sign is wrong, and the lower sign belongs to $-\pi\le y\le 0$, not $\pi\le y\le 2\pi$:

$$\int_0^\infty \sin(2h\cos y\cosh t)\,\mathrm{Ce}_{2n}(t,h^2)\,\mathrm{d}t = +\frac{\pi A_0^{2n}(h^2)}{2\,\mathrm{ce}_{2n}(\frac12\pi,h^2)}\left(\mathrm{ce}_{2n}(y,h^2)\mp\frac{2\,\mathrm{fe}_{2n}(y,h^2)}{\pi C_{2n}(h^2)}\right),$$

upper sign for $0\le y\le\pi$.

Evidence $h=0.8$, $y=2$: left side -0.1431816729, printed right side $+0.1431816729$ ($n=0$; the same for $2n=2,4$); the left side is even in y and fe_{2n} is odd; at $y=\pi+0.3$ the printed range gives ±2.05 against -0.4207.

Lean `integral_sin_cosh_modMathieuCe` (`DLMF/Mathieu/Modified/Integrals.lean`), admitted.

a.92 §28.31(iii), Asymptotic Behavior, second paragraph

Printed "For $\xi>0$, the functions $\mathrm{hc}_p^m(z,-\xi)$, $\mathrm{hs}_p^m(z,-\xi)$ behave asymptotically as multiples of $\exp\left(\frac14\xi\cos(2z)\right)(\cos z)^{-p-2}$ as $z\to\frac12\pi\pm\mathrm{i}\infty$. All other periodic solutions behave as multiples of $\exp\left(-\frac14\xi\cos(2z)\right)(\cos z)^{p}$."

Correct the powers p and $-p-2$ are swapped (writing $\sin z$ for $\cos z$ changes only the constant multiple, since $\sin z/\cos z\to\pm\mathrm{i}$ as $z\to\frac12\pi\pm\mathrm{i}\infty$): for $\xi>0$, $\mathrm{hc}_p^m(z,-\xi)$ and $\mathrm{hs}_p^m(z,-\xi)$ behave like multiples of

$$\exp\left(\tfrac14\xi\cos 2z\right)(\sin z)^{p}\qquad\text{as } z\to\tfrac12\pi\pm\mathrm{i}\infty,$$

and the other periodic solutions like $\exp\left(-\frac14\xi\cos 2z\right)(\sin z)^{-p-2}$.

Evidence DLMF (28.31.19)–(28.31.20) and the first paragraph; numerically the ratio to the corrected form converges (-0.416, -0.472, -0.480 for hc_2^0 at $y=2,3,4$), while the ratio to the printed form grows like $\mathrm{e}^{(2p+2)y}$.

Lean `paraboloidal_asymptotic_neg` (`DLMF/Mathieu/Hill/Ince.lean`), admitted.

a.93 (30.11.6), $j=1,2$

Printed

$$S_n^{m(j)}(z,\gamma)=\begin{cases}\psi_n^{(j)}(\gamma z)+O\left(z^{-2}\mathrm{e}^{|\Im z|}\right), & j=1,2,\\ \psi_n^{(j)}(\gamma z)\left(1+O\left(z^{-1}\right)\right), & j=3,4.\end{cases}$$

Correct the error term for $j=1,2$ must be $O\left(z^{-2}\mathrm{e}^{|\Im(\gamma z)|}\right)$, not $O\left(z^{-2}\mathrm{e}^{|\Im z|}\right)$ (false for $\gamma>1$):

$$S_n^{m(j)}(z,\gamma)=\psi_n^{(j)}(\gamma z)+O\left(z^{-2}\mathrm{e}^{|\Im(\gamma z)|}\right),\qquad j=1,2.$$

Evidence $m=n=0$, $\gamma=3$, $z=x(1+0.5\mathrm{i})$, $x=5,\dots,40$: $|\mathrm{err}|\,|z|^2\mathrm{e}^{-|\Im z|}=13.8,\dots,2.3\cdot10^{16}$, while $|\mathrm{err}|\,|z|^2\mathrm{e}^{-|\Im(\gamma z)|}=0.093,\dots,0.099$.

Lean `spheroidalRadial_isBigO_one_two` (`DLMF/Spheroidal/Radial.lean`), admitted, real $\gamma > 0$.

a.94 (31.10.8)

Printed

$$\sin^2\theta\left(\frac{\partial^2\mathcal{K}}{\partial\theta^2}+\left((1-2\gamma)\tan\theta+2(\delta+\epsilon-\tfrac12)\cot\theta\right)\frac{\partial\mathcal{K}}{\partial\theta}-4\alpha\beta\mathcal{K}\right)$$
$$+\frac{\partial^2\mathcal{K}}{\partial\phi^2}+\left((1-2\delta)\cot\phi-(1-2\epsilon)\tan\phi\right)\frac{\partial\mathcal{K}}{\partial\phi}=0.$$

Correct the coefficient of $\partial\mathcal{K}/\partial\phi$ must be $(2\delta-1)\cot\phi-(2\epsilon-1)\tan\phi$:

$$\sin^2\theta\left(\frac{\partial^2\mathcal{K}}{\partial\theta^2}+\left((1-2\gamma)\tan\theta+2(\delta+\epsilon-\tfrac12)\cot\theta\right)\frac{\partial\mathcal{K}}{\partial\theta}-4\alpha\beta\mathcal{K}\right)$$
$$+\frac{\partial^2\mathcal{K}}{\partial\phi^2}+\left((2\delta-1)\cot\phi-(2\epsilon-1)\tan\phi\right)\frac{\partial\mathcal{K}}{\partial\phi}=0.$$

Evidence exact symbolic comparison with $D_z - D_t$ on test polynomials (constant ratio only for the corrected coefficient); the kernel (31.10.10) has residual 10^{-31}, against 1.18 as printed.

Lean `heun_kernel_typeI_angles` (`DLMF/Heun/Integral.lean`), admitted.

a.95 (31.10.17)

Printed

$$u=\frac{(stz)^{1/2}}{a},\qquad v=\left(\frac{(s-1)(t-1)(z-1)}{1-a}\right)^{1/2},\qquad w=\mathrm{i}\left(\frac{(s-a)(t-a)(z-a)}{a(1-a)}\right)^{1/2}.$$

Correct u must be $(stz/a)^{1/2}$, not $(stz)^{1/2}/a$:

$$u=\left(\frac{stz}{a}\right)^{1/2};\qquad\text{then}\quad u^2+v^2+w^2=s+t+z-1-a.$$

Evidence DLMF (31.10.14) against DLMF (31.10.18) on test polynomials (sympy): constant ratio only with $u^2 = stz/a$.

Lean `heun_kernel_typeII_uvw` (`DLMF/Heun/Integral.lean`), admitted.

a.96 (31.10.22)–(31.10.23)

Printed

$$\mathcal{K}(r,\theta,\phi)=r^m\sin^{2p}\theta\,P\left\{\begin{array}{cccc}0&1&\infty&\\0&0&a&\cos^2\theta\\\frac12(3-\gamma)&c&b&\end{array}\right\}P\left\{\begin{array}{cccc}0&1&\infty&\\0&0&a'&\cos^2\phi\\1-\epsilon&1-\delta&b'&\end{array}\right\},$$

with

$$m^2+2(\alpha+\beta)m-\sigma_1=0,\qquad p^2+(\alpha+\beta-\gamma-\tfrac12)p-\tfrac14\sigma_2=0,$$
$$a+b=2(\alpha+\beta+p)-1,\qquad ab=p^2-p(1-\alpha-\beta)-\tfrac14\sigma_1,\qquad c=\gamma-\tfrac12-2(\alpha+\beta+p),$$
$$a'+b'=\delta+\epsilon-1,\qquad a'b'=-\tfrac14\sigma_2.$$

The printed $\cos^2\theta$ P-symbol violates Fuchs's relation (its exponents sum to $\gamma/2$).

Correct exponent $1-\gamma$ at 0 (not $\frac12(3-\gamma)$), and

$$p^2+(\alpha+\beta-\gamma)p-\tfrac14\sigma_2=0,\qquad a+b=\alpha+\beta+2p,$$
$$ab=p^2+(\alpha+\beta)p-\tfrac14\sigma_1,\qquad c=\gamma-\alpha-\beta-2p.$$

The m-equation and the a', b' relations are right.

Evidence separation of DLMF (31.10.21) by hand; the corrected product kernel leaves a residual of $2 \cdot 10^{-31}$ in (31.10.21), the printed one 8.2.

Lean `heun_kernel_typeII_separable` (`DLMF/Heun/Integral.lean`), admitted.

a.97 §31.10, Example 2 ((31.10.11))

Printed Fuchs–Frobenius solutions

$$W_m(z) = \tilde{\kappa}_m z^{-\alpha}\,\mathrm{Hl}\left(1/a, q_m; \alpha, \alpha-\gamma+1, \alpha-\beta+1, \delta; 1/z\right)$$

represented in terms of Heun functions by DLMF (31.10.1) with the kernel (31.10.11).

Correct $z^{-\alpha}\,\mathrm{Hl}(1/a, q_m; \alpha, \alpha-\gamma+1, \alpha-\beta+1, \delta; 1/z)$ does not solve Heun's equation; the accessory parameter must be

$$\frac{q_m}{a} - \alpha(\beta-\epsilon) - \frac{\alpha}{a}(\beta-\delta),$$

as in the corrected DLMF (31.3.10) (DLMF erratum [2], Karney 2022), which Example 2 was not brought in line with.

Evidence $a = 0.4$, $q = 0.37$, $(\alpha, \beta, \gamma, \delta) = (0.6, 1.9, 1.3, 0.7)$, $z = 5$: residual $2.2 \cdot 10^{-3}$ as printed, $1.5 \cdot 10^{-32}$ corrected.

Lean only the kernel equation is acquired (`heun_kernel_example`$_2$`_pde`, `DLMF/Heun/Integral.lean`); the representation is not.

3 (b) Project transcription defects that were fixed

b.1 (4.19.3), (4.19.4), (4.22.5)

Defect Mistaken tags: the tangent series was tagged E4, the cosecant series E5. The statements were unchanged.

Lean `Complex.hasSum_pi_mul_tan_bernoulli`, `Complex.hasSum_pi_mul_div_sin_bernoulli` (`DLMFMathlib/NumberTheory/BernoulliTaylor.lean`), proved.

Found 2026-10-24 (`DLMF4Acquisition.md`).

b.2 (4.38.5)

Defect The earlier Lean statement asserted unordered `HasSum` of $z^{2n+1}/(2n+1)$ to $\operatorname{arctanh} z$ on the closed disk $|z| \le 1$, $z \neq \pm 1$. On the unit circle this is false: at $z = \mathrm{i}$ the term norms are $1/(2n+1)$, so the series is not absolutely summable, and in $\mathbb{C}$ unordered summability means absolute summability. DLMF's statement ($|z| \le 1$, $z \neq \pm 1$) is right with ordinary convergence.

Fix Unordered `HasSum` for $|z| < 1$, plus natural-order convergence of the partial sums on DLMF's full domain.

Lean `DLMF.Elementary.hasSum_arctanh`, `DLMF.Elementary.tendsto_arctanh_series` (`DLMF/Elementary/Hyperbolic.lean`), both proved (from the boundary logarithm series `logBoundary_odd`).

Found Found and fixed 2026-10-25.

b.3 (4.38.6)

Defect Same defect for the series of $1/((2n+1)z^{2n+1})$ for $|z| \ge 1$, $\Im z \neq 0$ ($z = \mathrm{i}$ again gives term norms $1/(2n+1)$).

Fix Unordered `HasSum` for $|z| > 1$, natural-order convergence for $|z| \ge 1$, $\Im z \neq 0$, with the sum $\operatorname{arctanh} z \mp \mathrm{i}\pi/2$ for $\Im z \gtrless 0$.

Lean `DLMF.Elementary.hasSum_arctanh_large`, `DLMF.Elementary.tendsto_arctanh_large_series` (`DLMF/Elementary/Hyperbolic.lean`), both proved.

Found Found and fixed 2026-10-25.

b.4 (4.38.15)

Defect The earlier full-set statements dropped the correlation between square-root choices and inverse branches ($u = v = \mathrm{i}$: the right side contains $w = 0$).

Fix Shared roots and paired sinh/cosh equations.

Lean `arcsinh_values_add`, `arcsinh_values_sub` and Lean counterexamples to the former form (`DLMF/Elementary/Hyperbolic.lean`), proved.

Found Found and fixed 2026-10-24 (`REVIEW.md`, Chapters 4, 16 and 17 rounds).

b.5 (4.38.18)

Defect The same defect (the former sets fail at $u = \mathrm{i}$, $v = 1$).

Lean `arcsinh_arccosh_values_add`, `arcsinh_arccosh_values_sub`, counterexample `arcsinh_arccosh_values_unrestricted_counterexample`, proved.

Found Found and fixed 2026-10-24.

b.6 (5.9.4)

Defect An integral binder extended over the following sum, so the statement asserted $\Gamma(z) = 0$.

Fix Parenthesizing.

Lean `Gamma_incomplete_sum` (`DLMF/Gamma/IntegralRepresentations.lean`), proved.

Found 2026-10-15 (`REVIEW.md`, "Gamma pass from Temme's Chapter 3").

b.7 §§8.9–8.17 continued fractions (borderline (b)/(c))

Defect Draft finite continued fractions used nested totalized division.

Fix Replaced by ratios of continuants.

Found 2026-10-19 (`DLMF678Acquisition.md`).

b.8 (9.6.11)

Defect Wrong bracket (copied from the shape of DLMF (9.6.13)), false at $z = 1$.

Lean `besselJ_third_eq_airy_complex` (`DLMF/Airy/Relations.lean`), proved.

Found Found 2026-10-19, fixed 2026-10-20 (`REVIEW.md`, "Project-wide admission triage").

b.9 (9.7.17) (borderline (b)/(c): caused by $0^{-1} = 0$)

Defect The error multiplier was 0 on $|\arg z| = 2\pi/3$, so the bound asserted $\|R_N\| \le 0$.

Lean `airyErrorMultiplier`, `airy_ai_complex_error` (proved), `airy_aiDeriv_complex_error` (admitted) (`DLMF/Airy/Asymptotics.lean`).

Found 2026-10-21 (`REVIEW.md`, "Round 8").

b.10 (9.8.22), (9.8.23)

Defect The phase remainders were empty at $N = 0$.

Lean `airyPhase_asymptotic`, `airyDerivPhase_asymptotic` (`DLMF/Airy/ModulusPhase.lean`), proved.

Found 2026-10-19 (`REVIEW.md`, "Airy acquisition and first proof batch").

b.11 (9.13.23), (9.13.24)

Defect Odd m was allowed, but W_m is defined only for even m.

Lean `smirnovAiryU1_olver`, `smirnovAiryU2_olver` (`DLMF/Airy/Generalized.lean`), proved.

Found 2026-10-20 (`REVIEW.md`, "Project-wide admission triage").

b.12 (11.10.24)

Defect Wrong upper limit in `weberM1` (empty sum at $n = 1$).

Lean `weberM1`, `weberE_zero` (`DLMF/Struve/AngerWeber.lean`).

Found 2026-10-20 (same section as the preceding entry).

b.13 (11.11.3)

Defect An auto-bound implicit `k` in the G_k term made the statement quantify over it.

Fix Bracketing the summand.

Lean `weberE_large_argument` (`DLMF/Struve/Asymptotics.lean`), proved.

Found 2026-10-21 (`REVIEW.md`, "Round 6").

b.14 (12.10.11)

Defect Coefficient $\frac{3}{2}(s+1)$ instead of DLMF's $3(s+1)$ in `pcfOlverPair`.

Lean `pcfOlverU_spec`, `pcfOlverUV_initial` (`DLMF/ParabolicCylinder/Uniform.lean`), proved.

Found 2026-10-20 (`REVIEW.md`, "Project-wide admission triage").

b.15 (13.9.11)–(13.9.14)

Defect The sector-count formulas were misread as positive-zero counts ($(a, b) = (-1.3, -2.2)$ has no positive zeros).

Lean `kummerU_sectorZeros_gamma_pos`, `kummerU_sectorZeros_gamma_neg` (admitted), `kummerU_posZeros_ceil` (proved) (`DLMF/Confluent/Zeros.lean`).

Found 2026-10-20.

b.16 (13.9.15)

Defect `kummerU_negZeros_none` was tagged 13.9.15 but concerns the negative axis, and was false for nonpositive integer a.

Fix The tag was removed and DLMF (13.9.15) is proved as `kummerU_posZeros_none_of_le` (`DLMF/Confluent/Zeros.lean`).

Found 2026-10-16 and 2026-10-20 (`REVIEW.md`, "Admission pass after the Temme and Duverney work" and "Project-wide admission triage").

b.17 (14.15.21)–(14.15.22)

Defect `LegendreYRelation` admitted two roots ($\alpha = 0.6$, $x = 0.9$: $y = 0.177237$ and -0.116411).

Fix Sign conditions.

Lean `legendreDegreeY_spec` (`DLMF/Legendre/UniformCoordinates.lean`), proved.

Found 2026-10-22 (`REVIEW.md`, "Round 8").

b.18 §14.20(viii) (borderline (b)/(c))

Defect `conicalEta_mapping` allowed $\alpha = 0$, where $\eta \to \pi^2$ rather than ∞.

Fix Now $0 < \alpha$. Proved (`DLMF/Legendre/ConicalAsymptotics.lean`).

Found 2026-10-20.

b.19 (16.11.4) (coefficients $e_{k,m}$; definition `genAsymptoticEkm`, tagged 16.11.E5)

Defect `∏ i, a i - b j` parsed as $(\prod_i a_i) - b_j$; DLMF means $\prod_i (a_i - b_j)$, and likewise in the denominator.

Fix Parenthesizing.

Lean `genAsymptoticC_limit`, `genAsymptoticC_recurrence` (`DLMF/GeneralizedHypergeometric/Asymptotics.lean`), admitted.

Found 2026-10-25 (`REVIEW.md`, "Parallel proofs in Chapters 4, 16 and 17 (2026-10-24, fourth round)").

b.20 (19.8.6)

Defect The summand `agmSeqC 1 g₀ n ^ 2` used the real square root $c_n = \sqrt{a_n^2 - g_n^2}$, which is 0 at $n = 0$ when $k^2 < 0$ (then $g_0 = k' > a_0 = 1$), whereas DLMF has $c_0^2 = k^2$. The first identity was false for $k^2 < 0$ (mpmath at $k^2 = -1$: 0.04305 against -0.45695).

Fix The summand is $a_n^2 - g_n^2$ (DLMF (19.8.2)).

Lean `compEllIntEk_eq_agm` (`DLMF/EllipticIntegrals/Quadratic.lean`), proved for all real $k^2 < 1$.

Found 2026-10-25 (proof round 1).

b.21 (19.22.9), (19.22.10), (19.22.12), (19.22.14)

Defect The AGM sequences `agmA`, `agmG` (`DLMF/EllipticIntegrals/SymmetricQuadratic.lean`) took index $n + 1$ from Mathlib's `NNReal.agmSequences (n + 1)`, which already holds the $(n + 2)$-th means, so a_1 and g_1 were skipped and all four series were false ($a_0 = 2$, $g_0 = 1$: $(4/\pi)R_G = 1.541964$ against 1.714838).

Fix Index $n + 1$ uses `agmSequences n`.

Lean `symEllIntRG_zero_agm`, `symEllIntRD_zero_agm` proved; (19.22.12), (19.22.14) admitted.

Found 2026-10-25 (proof round 1).

b.22 §20.2(ii)

Defect The earlier `jacobiThetaOneTau_div_sin_analytic` asked, for each fixed z, only for some analytic function of τ agreeing with $\theta_1(z|\tau)/\sin z$ where $\sin z \neq 0$ (and likewise $\theta_2/\cos z$). At the zeros of $\sin z$ this is vacuous, so the statement was trivially true and missed DLMF's meaning.

Fix Restated (user decision): one function $F(z, \tau)$, entire in z, analytic in τ on $\Im\tau > 0$, equal to the quotient where $\sin z \neq 0$, so that at the zeros it is the removable-singularity value. Proved (F is the Fourier series DLMF (20.2.1) divided termwise by $\sin z$, via the Dirichlet kernel; $G(z, \tau) = F(\pi/2 - z, \tau)$).

Lean `DLMF/Theta/Basic.lean`.

Found 2026-10-25 (proof round 1), restated and proved the same day.

b.23 (28.2.33) and §28.4(i)

Defect The acquired statements ranged over all $n \in \mathbb{N}$, including $n = 0$, but DLMF has no se_0: DLMF (28.4.3)–DLMF (28.4.4) index the functions as se_{2n+1}, se_{2n+2}, $n = 0, 1, 2, \ldots$. For the Lean junk value se_0 (at $q = 0$ a positive multiple of x) the orthogonality DLMF (28.2.33) fails, and the sine series of §28.4(i) diverges off the real axis.

Fix The hypothesis $n \neq 0$ (2026-10-26).

Lean Counterexample `p1a1_integral_mathieuCe_mul_mathieuSe_zero_ne` (proved); `integral_mathieuCe_mul_mathieuSe` (`DLMF/Mathieu/Properties.lean`), now proved; second conjunct of `mathieu_fourier_locallyUniform` (`DLMF/Mathieu/FourierSeries.lean`), admitted.

Found 2026-10-26 (proof round 1 for Chapters 28–31).

4 (c) Formalization notes

4.1 Totalization, removable points and nonvanishing hypotheses

c.1 (4.18.3), (4.32.1), (4.40.9)

Note Removable values ($\sin x/x$ and $\sinh x/x$ at 0; the value 4 at $a = 0$) are stated explicitly.

Lean `cos_le_sinc_le_one` (proved), `cosh_le_sinhc_cube` and `integral_exp_div_cosh_sq` (admitted).

Source `DLMF4Acquisition.md`.

c.2 (4.24.15)

Note The finite quotient needs $1 \mp uv \neq 0$.

Lean `arctan_general_add`, proved.

Source `DLMF4Acquisition.md`.

c.3 (5.9.7)

Note At $x = 0$ the totalized right side vanishes; restated as $\pi/(2\Gamma(1-x)\cos(\pi x/2))$.

Lean `Gamma_sin`, proved.

Found 2026-10-07 (`REVIEW.md`, "Statement defects repaired on 2026-10-07").

c.4 (5.12.8)

Note Mathlib's `betaIntegral` is the literal integral; the right side is written with Gamma functions.

Lean `beta_vertical`, proved.

Found 2026-10-17 (`REVIEW.md`, "Remaining Gamma admissions, second round").

c.5 (8.10.8) (borderline (a)/(c))

Note The integral needs $a > 0$; $a = 0$ as a limit.

Lean `incompleteGamma_positive_exponential_bounds`, `incompleteGamma_positive_exponential_limit`, proved.

Source `DLMF678Acquisition.md`.

c.6 (11.4.18)

Note $\nu + \frac{1}{2} \notin -\mathbb{N}$ added (removable term totalized to 0).

Lean `struveH_bessel_series_18`, proved.

Found 2026-10-20.

c.7 §11.10, Anger scaled coefficients

Note Removable limits at $\lambda = 1$ stated as limits.

Source `DLMF111214Acquisition.md`.

c.8 (13.6.11)

Note $\Gamma(0) = 0$ in Lean; nonpositive integer ν ($\nu - n$) excluded.

Lean `kummerM_besselI_sum`, `kummerM_besselI_sum_neg`, proved.

Found 2026-10-07.

c.9 (13.16.8) (borderline (a)/(c))

Note The Bochner integral also needs $\Re(-\mu - \kappa) + \frac{1}{2} > 0$.

Lean `whittakerW_integral_besselK`, proved.

Found 2026-10-08 (`REVIEW.md`, "Further proofs and a statement correction").

c.10 (13.17.1), (13.17.3)

Note Denominator $\neq 0$ added ($x/0 = 0$ at the poles).

Lean `whittakerM_continued_fraction_limit`, `whittakerW_continued_fraction_limit`, proved.

Found 2026-10-21.

c.11 (13.23.3), (13.23.8), (13.23.9), (13.23.10), (13.23.12)

Note Ordinary M divided by $\Gamma(-n) = 0$ replaced by the regularized Whittaker M.

Lean `mellin_whittakerM`, `cosine_whittakerM`, `bessel_whittakerM`, `bessel_whittakerM_W`, `bessel_whittakerW_M`, proved.

Found 2026-10-18 (`REVIEW.md`, "Whittaker corrections and the first prioritized proof batch").

c.12 (14.12.4) (borderline (a)/(c))

Note $\Re(\mu + \nu) > -1$ for convergence.

Lean `associatedP_cosh_integral`, proved.

Source `DLMF111214Acquisition.md`.

c.13 (14.15.23) (borderline (a)/(c))

Note $X_0 = 0$ is degenerate; now $0 < c \to 0 < X_c$.

Lean `cylinderEnvelopeCross_spec`, admitted.

Found 2026-10-22 (`REVIEW.md`, "Round 9 statement correction").

c.14 (14.20.10)

Note The singular endpoint $\theta = \pi$ excluded.

Lean `conical_power_series`, proved.

Source `DLMF111214Acquisition.md`.

c.15 (17.6.28)

Note Gamma poles excluded, with a parameter-limit statement.

Lean `heine_jackson_integral`, `heine_jackson_parameter_limit`, admitted.

Source `DLMF17Acquisition.md`.

c.16 (17.13.3)

Note Removable-limit values of a Gamma quotient, with a limit statement and a literal regular-strip statement.

Lean `DLMF/QHypergeometric/Integrals.lean`.

Source `DLMF17Acquisition.md`.

c.17 (20.7.11), (20.7.12), (20.7.25), (20.7.34), (20.8.1), (20.11.9)

Note Where DLMF divides by a theta function, a nonvanishing hypothesis is added; with Lean's $x/0 = 0$ the printed quotients are false at the zeros.

Lean `jacobiTheta_reduction_two_mul`, `jacobiTheta_reduction_two_tau`, `hasDerivAt_jacobiThetaTwoTau_div_four`, the three (20.7.34) forms, `jacobiTheta_watson_expansion` (`DLMF/Theta/Identities.lean`, admitted); `combinedTheta_symmetry` (`DLMF/Theta/Generalizations.lean`, proved).

Found 2026-10-25.

c.18 (20.10.3)

Note The right side

$$(1-2^{1-s})\pi^{-s/2}\,\Gamma\!\left(\tfrac{1}{2}s\right)\zeta(s)$$

is removable at $s=1$, where Lean's totalized $\zeta(1)$ would make it 0. Stated for $s\neq 1$, with the value $\ln 2$ at $s=1$ separately.

Evidence mpmath at $s=0.6+0.2\mathrm{i}$, 1.5, 3, $2.5+0.3\mathrm{i}$ and $s=1$.

Lean `mellin_one_sub_jacobiThetaFourNullTau`, `mellin_one_sub_jacobiThetaFourNullTau_one` (`DLMF/Theta/Integrals.lean`), admitted.

Found 2026-10-25.

4.2 Branches and multivalued sets

c.19 (19.16.11), (19.21.4), (19.21.5)

Note Unprinted branch conditions made explicit: $\operatorname{ph}\lambda+\operatorname{ph}z_j\in(-\pi,\pi)$ for the homogeneity (19.16.11), and the sign of $\Im z$ (`SignType.sign z.im`) in (19.21.4)–(19.21.5).

Lean `symEllIntR_mul` (proved) and the 19.21 connection formulas (`DLMF/EllipticIntegrals/SymmetricConnection.lean`).

Found 2026-10-25 (`DLMF19Acquisition.md`).

c.20 (4.5.3), (4.5.14) (borderline (a)/(c))

Note Strict below the exact nonzero root, weak at it; the decimal cutoff is excluded as numerical.

Lean `abs_log_one_sub_lt`, `abs_log_one_sub_le`, `exp_neg_lt_linear`, `exp_neg_le_linear`, proved.

Source `DLMF4Acquisition.md`.

c.21 (4.8.5)

Note $n\operatorname{Ln}z$ alone misses branches; all integer shifts of $n\log z$ are used.

Lean `generalLogValues_zpow`, proved.

Source `DLMF4Acquisition.md`.

c.22 (4.24.13)

Note The literal independent-root set equality is false ($u=v=1$); the branch-compatible identity is proved.

Lean `arcsin_general_add`, proved.

c.23 (4.24.16)

Note Square-root signs correlated with the inverse values.

Lean `arcsin_arccos_general_add`, proved.

c.24 (8.7.2)

Note The segment from x to $x+y$ must avoid the cut (otherwise it fails, e.g. at $a=0.3$, $x=-1+0.1\mathrm{i}$, $y=-0.2\mathrm{i}$).

Lean `incompleteGamma_increment_series`, proved.

Found 2026-10-22 (`REVIEW.md`, "Round 8").

c.25 (9.6.11), (9.6.13), (9.6.15), (9.6.17), (9.6.19)

Note $(3/z)^{1/2}$ written as $\sqrt{3}\,z^{-1/2}$ to match the logarithmic ζ on the left.

Lean The `..._third_eq_airy_complex` family (`DLMF/Airy/Relations.lean`), proved.

Found 2026-10-20.

c.26 (9.13.27), (9.13.31), (9.13.32) (borderline (a)/(c))

Note The ray integrals diverge at 0 for positive integer p; a Hadamard finite part is used and the homogeneous equations are asserted for $p \le 0$.

Lean Proved (`DLMF/Airy/GeneralizedContours.lean`).

Source `AiryAcquisition.md`.

c.27 (12.13.5)

Note Endpoint hypothesis for $t = \pi/4$ added; no counterexample claimed.

Lean `pcfU_rotational_addition`, admitted.

Source `DLMF111214Acquisition.md`.

4.3 Implicit conventions made explicit

c.28 (21.2.6)

Note The hypothesis $\Omega^{\mathsf{T}} = \Omega$ is stated (DLMF's Ω is a Riemann matrix, hence symmetric).

Lean `riemannThetaChar_eq` (`DLMF/RiemannTheta/Basic.lean`), proved.

Found 2026-10-25.

c.29 (21.5.5)

Note $\mathbf{A}$ must be invertible over $\mathbb{Z}$ ($\mathbf{A} \in \mathrm{GL}_g(\mathbb{Z})$, so that $\mathbf{\Gamma}$ is an integer symplectic matrix), not merely $\det \mathbf{A} \neq 0$.

Lean `riemannTheta_unimodular` (`DLMF/RiemannTheta/Modular.lean`), proved.

Found 2026-10-25.

c.30 (21.6.3), (21.6.4)

Note The representatives of $\mathcal{K}$ are taken in $\mathbb{Z}^{g\times h}\mathbf{T}$, as DLMF's definition (21.6.1) says; with arbitrary lifts modulo integer matrices the identity fails ($g = 1$, $\mathbf{T} = \left[\begin{smallmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{smallmatrix}\right]$).

Lean `riemann_identity`, `riemann_identity_char` (`DLMF/RiemannTheta/Products.lean`), admitted.

Found 2026-10-25.

4.4 Chapters 24–27

c.31 Chapter 25 removable points

Note (25.4.2) ($s \neq 2, 3, \dots$), (25.4.5) ($s \neq -1, -2, \dots$), (25.5.13) ($s \neq 0$), (25.8.3) ($s \neq 0, -1, \dots$), (25.8.6) ($z \neq 0$) and (25.10.4) ($\cos\sqrt{2\pi t} \neq 0$) carry hypotheses because Lean's Γ vanishes at its poles, $\cot 0 = 0$ and $\zeta(1)$ is a junk value; (25.2.3), (25.5.3), (25.5.4), (25.5.10) and (25.5.21) are stated multiplied by $1 - 2^{1-s}$, which is stronger than DLMF.

Lean `DLMF/Zeta`.

Source `DLMF25Acquisition.md`, part A.

c.32 (25.10.2)

Note $\theta(t)$ uses the principal arg, as DLMF prescribes; it differs from the continuous Riemann–Siegel θ by multiples of 2π.

Lean `riemannSiegelTheta` (`DLMF/Zeta/Zeros.lean`), definition.

c.33 Chapter 25 implicit conditions

Note Made explicit by the acquisition (not independently re-verified): $s \neq -m$ in (25.11.16); $s \neq 0$ in (25.11.17) and (25.11.41); $s \neq 1$ in (25.11.31) and (25.11.35); $k \geq 2$ in (25.15.3), (25.15.6) and (25.15.10); $z \neq 0$ and $z \notin [1, \infty)$ in (25.12.12) and (25.14.3_1).

Source `DLMF25Acquisition.md`, part B.

c.34 (25.11.10)

Note At $s = 0, -1, -2, \ldots$ the term with $n + s = 1$ is DLMF's limit of $(s)_n \zeta(n + s)$, whereas Lean's `poch s n · ζ(1)` is 0; at $s = 0$, $a = \frac{1}{2}$ the Lean series sums to $-\frac{1}{2}$ while $\zeta(0, \frac{1}{2}) = 0$. The hypothesis $s \neq 0, -1, -2, \ldots$ was added.

Lean `r25b_hasSum_hurwitzZetaComplex_taylor_counterexample` (proved), `hasSum_hurwitzZetaComplex_taylor` (`DLMF/Zeta/Hurwitz.lean`), admitted.

Found 2026-10-25 (proof round 2).

c.35 (26.8.5), (26.8.27), (26.8.33)

Note The hypothesis $k \leq n$ comes from Lean's truncated natural subtraction and from the left side's domain; DLMF is right as printed (independent check 2026-10-25).

Lean `DLMF/Combinatorics/Stirling.lean`.

4.5 Chapters 28–31

c.36 (28.6.16)

Note The continued fraction holds for $n \geq 1$; for $n = 0$ the first numerator is $-2q^2$:

$$a_0 = \cfrac{-2q^2}{4 - a - \cfrac{q^2}{16 - a - \cdots}}.$$

Both forms are acquired.

Lean `mathieuA_even_contFrac` (`DLMF/Mathieu/SmallQ.lean`).

c.37 (28.5.5), $n = 0$

Note The normalization forces $C_0(q) \to \infty$ as $q \to 0$, so fe_0 with this normalization is discontinuous at $q = 0$, where (28.5.7) sets $\mathrm{fe}_0(z, 0) = z$ by convention; the convention is built into `mathieuFe` (`DLMF/Mathieu/Basic.lean`).

c.38 (28.6.12)–(28.6.13)

Note No erratum. The q^6 denominator is 92935987200000 as printed (grouped `9293 59872 00000`); an acquisition agent's first report misread the grouping.

c.39 §29.8

Note The kernels use the Ferrers function P_ν at arguments outside $[-1, 1]$; the analytic continuation $F(\nu + 1, -\nu; 1; \frac{1}{2} - \frac{1}{2}x)$, i.e. the Legendre P_ν, is meant, and the Lean statements use `DLMF.Legendre.legendreP`. Notation, not an erratum.

c.40 (31.14.3)–(31.14.5)

Note The normal form needs distinct singular points, implicit in DLMF; with coinciding a_j the Lean junk value $1/(a_j - a_k) = 0$ breaks it. The hypothesis `Function.Injective a` was added on 2026-10-26.

Lean Counterexample `p1h_fuchsian_normalForm_counterexample` (proved); `fuchsian_normalForm` (`DLMF/Heun/Fuchsian.lean`), admitted.

c.41 Domain choices

Note (28.12.2), (28.12.8), (28.12.9) and §28.14 are stated for $\nu \notin \mathbb{Z}$ (they fail for integer ν under DLMF's own completion, e.g. $\lambda_1(-q) = b_1(q)$); (28.23.2)–(28.23.5) are stated with the condition $|\cosh z| > 1$ ($|\sinh z| > 1$) for every j.

4.6 DLMF's own published errata followed

c.42 (22.16.14)

Note The corrected form (DLMF erratum of 2010-07-08, Karney) is used.

Lean `Real.jacobiEpsilonk_eq_integral_sn` (`DLMF/JacobiElliptic/Related.lean`), admitted.

c.43 (23.18.7)

Note The V1.0.11 erratum (no coprimality condition on r) is followed.

Lean `dedekindSum` (`DLMF/Weierstrass/ModularFunctions.lean`), definition.

5 Outside the DLMF scope

Errors in the supporting sources, recorded in Lean docstrings or reviews:

s.1 AAR (4.1.11)

Note The right-hand denominator is printed a instead of c (DLMF (13.2.39) is right).

Lean `kummerM_kummer_transform` (`DLMF/Confluent/Basic.lean`), proved.

s.2 AAR Theorem 8.9.1(a)

Note The last denominator $(\alpha)_{2k}$ should be $(\alpha+1)_{2k}$ ($k = 1$, $\alpha = \frac{1}{2}$, $\beta = 1$ gives $8/15$ against the printed $8/3$).

Lean `nearly_poised_3F2` (`DLMF/Selberg/Applications.lean`), proved.

s.3 Beals–Wong, p. 407

Note The substitution $\varphi(x) = \psi(\mathrm{i}\,2^{-1/2}x)$ needs a scale k with $k^3 = -\frac{1}{2}$ (`REVIEW.md`, "Beals–Wong Painlevé II").

Checked and cleared (not errata). (11.5.8) (`struveH_barnes`; the earlier quadrature disagreement was truncation) and (11.11.16) (`angerA_turning_equiv`).

References

[1] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds., *NIST Digital Library of Mathematical Functions*, https://dlmf.nist.gov/, Release 1.2.8 of 2026-09-15.

[2] NIST Digital Library of Mathematical Functions, *Errata*, https://dlmf.nist.gov/errata/, Release 1.2.8 of 2026-09-15.

[3] G. E. Andrews, R. Askey, and R. Roy, *Special Functions*, Encyclopedia of Mathematics and its Applications 71, Cambridge University Press, Cambridge, 1999.

[4] R. Beals and R. Wong, *Special Functions and Orthogonal Polynomials*, Cambridge Studies in Advanced Mathematics 153, Cambridge University Press, Cambridge, 2016.

[5] B. C. Carlson and J. L. Gustafson, Asymptotic approximations for symmetric elliptic integrals, *SIAM J. Math. Anal.* **25** (1994), 288–303.

[6] G. Gasper and M. Rahman, *Basic Hypergeometric Series*, 2nd ed., Encyclopedia of Mathematics and its Applications 96, Cambridge University Press, Cambridge, 2004.